

Satellite Orbits and Trajectories

Definition of an Orbit and a Trajectory

Trajectory: It is a path traced by a moving body. The term ‘trajectory’, is associated with a path that is not periodically revisited. The path followed by a rocket on its way to the right position for a satellite launch (Figure 2.1) or the path followed by orbiting satellites when they move from an intermediate orbit to their final destined orbit (Figure 2.2) are examples of trajectories

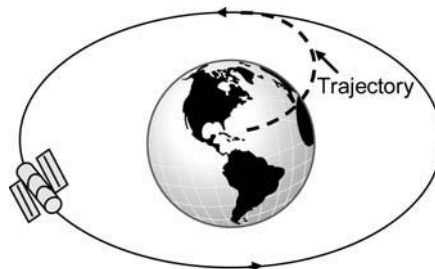


Figure 2.1 Example of trajectory – path followed by a rocket on its way during satellite launch

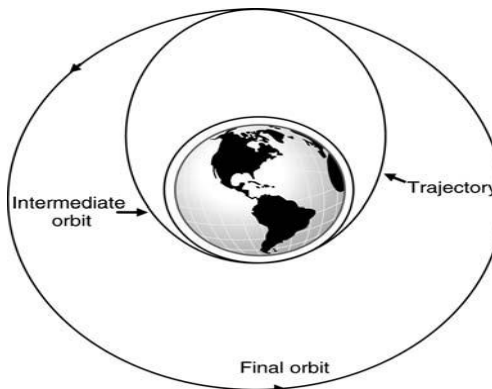


Figure 2.2 Example of trajectory – motion of a satellite from the intermediate orbit to the final orbit

Orbit : . The path followed by the motion of an artificial satellite around Earth is an orbit. Orbit is a trajectory that is periodically repeated. The motion of different planets of the solar system around the sun and the motion of artificial satellites around Earth (Figure 2.3) are examples of orbital motion.

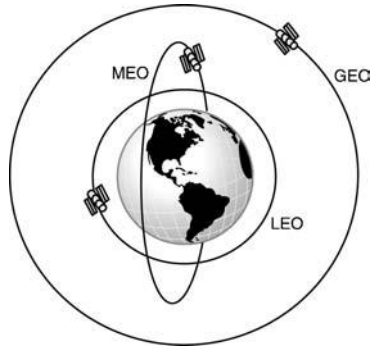


Figure 2.3 Example of orbital motion – satellites revolving around Earth

Orbiting Satellites – Basic Principles

The motion of natural and artificial satellites around Earth is governed by two forces.

Centripetal force directed towards the centre of the Earth due to the gravitational force of attraction of Earth

Centrifugal force that acts outwards from the centre of the Earth (Figure 2.4).

Centrifugal force: It is the force exerted during circular motion, by the moving object upon the other object around which it is moving. In the case of a satellite orbiting Earth, the satellite exerts a centrifugal force. The centrifugal force is simply the reaction force exerted by the satellite in a direction opposite to that of the centripetal force. This is in accordance with Newton's third law of motion, which states that for every action there is an equal and opposite reaction. This implies that there is a centrifugal acceleration acting outwards from the centre of the Earth due to the centripetal acceleration acting towards the centre of the Earth.

Centripetal force: The force that is causing the circular motion is the centripetal force. In the absence of this centripetal force, the satellite would have continued to move in a straight line at a constant speed after injection. The centripetal force directed at right angles to the satellite's velocity towards the centre of the Earth transforms the straight line motion to the circular or elliptical one, depending upon the satellite velocity.

Centripetal force leads to a corresponding acceleration called centripetal acceleration as it causes a change in the direction of the satellite's velocity vector.

The two forces can be explained from Newton's law of gravitation and Newton's second law of motion

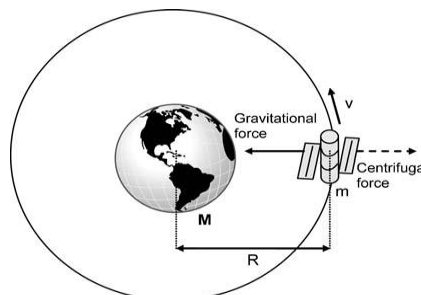


Figure 2.4 Gravitational force and the centrifugal force acting on bodies orbiting Earth

Newton's Law of Gravitation

According to Newton's law of gravitation, every particle irrespective of its mass attracts every other particle with a gravitational force whose magnitude is directly proportional to the product of the masses of the two particles and inversely proportional to the square of the distance between them and written as

$$F = \frac{Gm_1m_2}{r^2} \quad (2.1)$$

where

m_1, m_2 = masses of the two particles

r = distance between the two particles

G = gravitational constant = $6.67 \times 10^{-11} \text{ m}^3/\text{kg s}^2$

The force with which the particle with mass m_1 attracts the particle with mass m_2 equals the force with which particle with mass m_2 attracts the particle with mass m_1 . The forces are equal in magnitude but opposite in direction (Figure 2.5).

The acceleration, which is force per unit mass, experienced by the two particles, however, would depend upon their masses. A larger mass experiences lesser acceleration.

Newton also explained that, it is applicable to real objects as long as their sizes are small compared to the distance between them.

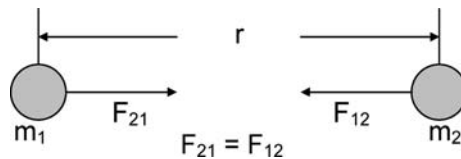


Figure 2.5 Newton's law of gravitation

Newton's Second Law of Motion

According to Newton's second law of motion, the force equals the product of mass and acceleration. In the case of a satellite orbiting Earth, if the orbiting velocity is v , then the acceleration, called centripetal acceleration, experienced by the satellite at a distance r from the centre of the Earth would be v^2/r . If the mass of satellite is m , it would experience a reaction force of mv^2/r . This is the centrifugal force directed outwards from the centre of the Earth and for a

satellite is equal in magnitude to the gravitational force.

If the satellite orbited Earth with a uniform velocity v , which would be the case when the satellite orbit is a circular one, then equating the two forces mentioned above would lead to an expression for the orbital velocity v as follows:

$$\frac{Gm_1m_2}{r^2} = \frac{m_2v^2}{r} \quad (2.2)$$

$$v = \sqrt{\left(\frac{Gm_1}{r}\right)} = \sqrt{\left(\frac{\mu}{r}\right)} \quad (2.3)$$

where

m_1 = mass of Earth

m_2 = mass of the satellite

$$\mu = Gm_1 = 3.986\,013 \times 10^5 \text{ km}^3/\text{s}^2 = 3.986\,013 \times 10^{14} \text{ N m}^2/\text{kg}$$

The orbital period in such a case can be computed from

$$T = \frac{2\pi r^{3/2}}{\sqrt{\mu}} \quad (2.4)$$

In the case of an elliptical orbit, the forces governing the motion of the satellite are the same. The velocity at any point on an elliptical orbit at a distance d from the centre of the Earth is given by the formula

$$v = \sqrt{\left[\mu \left(\frac{2}{d} - \frac{1}{a}\right)\right]} \quad (2.5)$$

where

a = semi-major axis of the elliptical orbit

The orbital period in the case of an elliptical orbit is given by

$$T = \frac{2\pi a^{3/2}}{\sqrt{\mu}} \quad (2.6)$$

The movement of a satellite in an orbit is governed by three Kepler's laws

Kepler's Laws

Kepler's First Law

The orbit of a satellite around Earth is elliptical with the centre of the Earth lying at one of the foci of the ellipse (Figure 2.6). The elliptical orbit is characterized by its semi-major axis a and eccentricity e .

Eccentricity is the ratio of the distance between the centre of the ellipse and either of its foci ($=ae$) to the semi-major axis of the ellipse a .

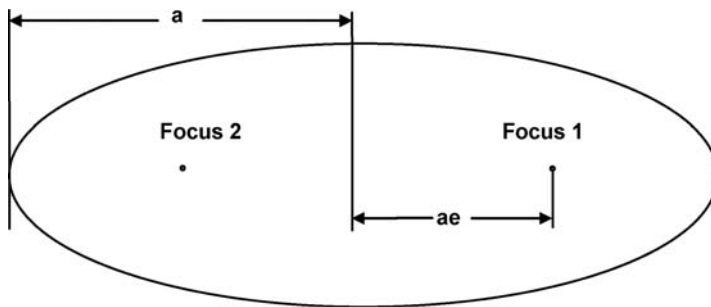


Figure 2.6 Kepler's first law

A circular orbit is a special case of an elliptical orbit where the foci merge together to give a single central point and the eccentricity becomes zero

Other important parameters of an elliptical satellite orbit include its **apogee** (farthest point of the orbit from the Earth's centre) and **perigee** (nearest point of the orbit from the Earth's centre) distances.

For any elliptical motion, the law of conservation of energy is valid at all points on the orbit. The law of conservation of energy states that energy can neither be created nor destroyed; it can only be transformed from one form to another. In the context of satellites, it means that the sum of the kinetic and the potential energy of a satellite always remain constant.

The value of this constant is equal to $-Gm_1m_2/(2a)$,
where

m_1 = mass of Earth

m_2 = mass of the satellite

a = semi-major axis of the orbit

The kinetic and potential energies of a satellite at any point at a distance r from the centre of the Earth are given by

$$\text{Kinetic energy} = \frac{1}{2}(m_2v^2) \quad (2.7)$$

$$\text{Potential energy} = -\frac{Gm_1m_2}{r} \quad (2.8)$$

Therefore,

$$\frac{1}{2}(m_2v^2) - \frac{Gm_1m_2}{r} = -\frac{Gm_1m_2}{2a} \quad (2.9)$$

$$v^2 = Gm_1 \left(\frac{2}{r} - \frac{1}{a} \right) \quad (2.10)$$

$$v = \sqrt{\left[\mu \left(\frac{2}{r} - \frac{1}{a} \right) \right]} \quad (2.11)$$

Kepler's Second Law

The line joining the satellite and the centre of the Earth sweeps out equal areas in the plane of the orbit in equal time intervals (Figure 2.7); i.e. the rate (dA/dt) at which it sweeps area A is constant.

The rate of change of the swept-out area is given by

$$\frac{dA}{dt} = \frac{\text{angular momentum of the satellite}}{2m} \quad (2.12)$$

where m is the mass of the satellite. Hence, Kepler's second law is also equivalent to the law of conservation of momentum, which implies that the angular momentum of the orbiting satellite given by the product of the radius vector and the component of linear momentum perpendicular to the radius vector is constant at all points on the orbit.

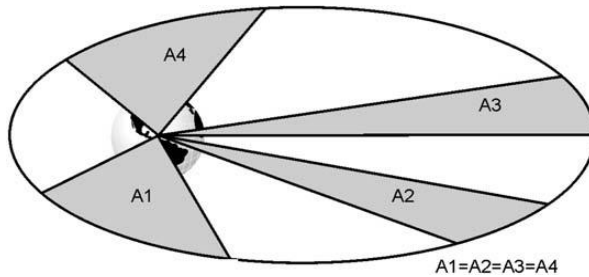


Figure 2.7 Kepler's second law

The angular momentum of the satellite of mass m is given by $mr^2\omega$, where ω is the angular velocity of the satellite.

$mr^2\omega$ ($m\omega r$) (r) $mv^r r$ remains constant.

Here v^r is the component of the satellite's velocity v in the direction perpendicular to the radius vector and is expressed as $v \cos \gamma$, where γ is the angle between the direction of motion of the satellite and the local horizontal, which is in the plane perpendicular to the radius vector r (Figure 2.8).

This leads to the conclusion that the product $rv \cos \gamma$ is constant. The product reduces to rv in the case of circular orbits and also at apogee and perigee points in the case of elliptical orbits due to angle γ becoming zero.

It is interesting to note here that the velocity component v^r is inversely proportional to the distance r . Qualitatively, this implies that the satellite is at its lowest speed at the apogee point and the highest speed at the perigee point. In other words, for any satellite in an elliptical orbit, the dot product of its velocity vector and the radius vector at all points is constant. Hence,

$$v_p r_p = v_a r_a = rv \cos \gamma \quad (2.13)$$

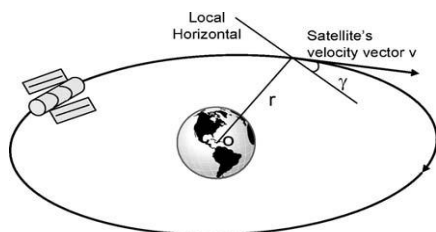


Figure 2.8 Satellite's position at any given time

where

v_p = velocity at the perigee point

r_p = perigee distance

v_a = velocity at the apogee point

r_a = apogee distance

v = satellite velocity at any point in the orbit

r = distance of the point

γ = angle between the direction of motion of the satellite and the local horizontal

Kepler's Third Law

According to the Kepler's third law, also known as the law of periods, the square of the time period of any satellite is proportional to the cube of the semi-major axis of its elliptical orbit.

The expression for the time period can be derived as follows. A circular orbit with radius r is assumed. Circular orbit is only a special case of an elliptical orbit with both the semi-major axis and semi-minor axis equal to the radius. Equating the gravitational force with the centrifugal force gives

$$\frac{Gm_1m_2}{r^2} = \frac{m_2v^2}{r} \quad (2.14)$$

Replacing v by ωr in the above equation gives

$$\frac{Gm_1m_2}{r^2} = \frac{m_2\omega^2r^2}{r} = m_2\omega^2r \quad (2.15)$$

which gives $\omega^2 = Gm_1/r^3$. Substituting $\omega = 2\pi/T$ gives

$$T^2 = \left(\frac{4\pi^2}{Gm_1} \right) r^3 \quad (2.16)$$

This can also be written as

$$T = \left(\frac{2\pi}{\sqrt{\mu}} \right) r^{3/2} \quad (2.17)$$

The above equation holds good for elliptical orbits provided r is replaced by the semi-major axis a . This gives the expression for the time period of an elliptical orbit as

$$T = \left(\frac{2\pi}{\sqrt{\mu}} \right) a^{3/2} \quad (2.18)$$

Orbital Parameters:

The satellite orbit, which in general is elliptical, is characterized by a number of parameters. These not only include the geometrical parameters of the orbit but also parameters that define its orientation with respect to Earth.

1. Ascending and descending nodes
2. Equinoxes
3. Solstices
4. Apogee
5. Perigee
6. Eccentricity
7. Semi-major axis
8. Right ascension of the ascending node
9. Inclination
10. Argument of the perigee
11. True anomaly of the satellite
12. Angles defining the direction of the satellite

Ascending and descending nodes. The satellite orbit cuts the equatorial plane at two points: the first, called the descending node (N1), where the satellite passes from the northern hemisphere to the southern hemisphere, and the second, called the ascending node (N2), where the satellite passes from the southern hemisphere to the northern hemisphere (Figure 2.9).

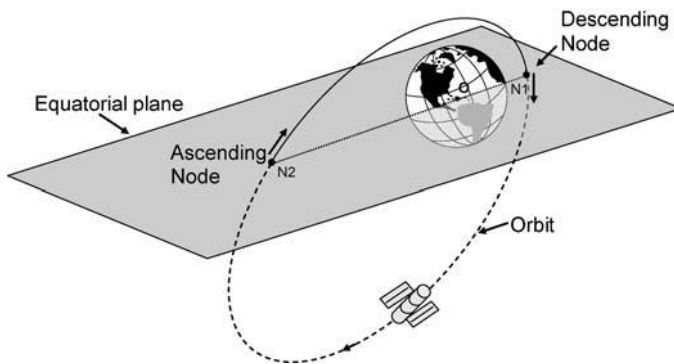


Figure 2.9 Ascending and descending nodes

Equinoxes. The Earth's equatorial plane is tilted with respect to the direction of the Sun. As the Earth orbits the Sun throughout the year, the angle between the Sun-Earth line and the equatorial plane changes continuously. This angle is called the solar declination angle, and it varies in a sinusoidal pattern over a period of 365 days.

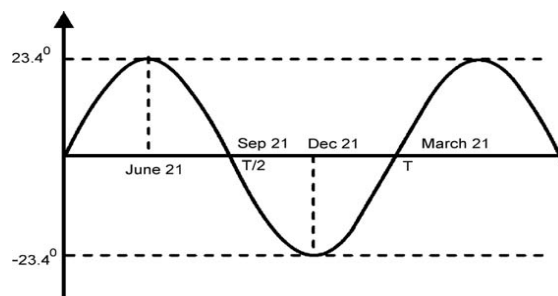


Figure 2.10 Yearly variation of angular inclination of Earth with the sun

The sinusoidal variation of the angle of inclination is defined by

$$\text{Inclination angle (in degrees)} = 23.4 \sin \left(\frac{2\pi t}{T} \right) \quad (2.19)$$

where T is 365 days. This expression indicates that the inclination angle is zero for $t = T/2$ and T . This is observed to occur on 20-21 March, called the spring equinox, and 22-23 September, called the autumn equinox.

The two equinoxes are understandably spaced 6 months apart. During the equinoxes, it can be seen that the equatorial plane of Earth will be aligned with the direction of the sun.

The line of intersection of the Earth's equatorial plane and the Earth's orbital plane that passes through the centre of the Earth is known as the line of equinoxes.

The direction of this line with respect to the direction of the sun on 20-21 March determines a point at infinity called the vernal equinox (Υ) (Figure 2.11).

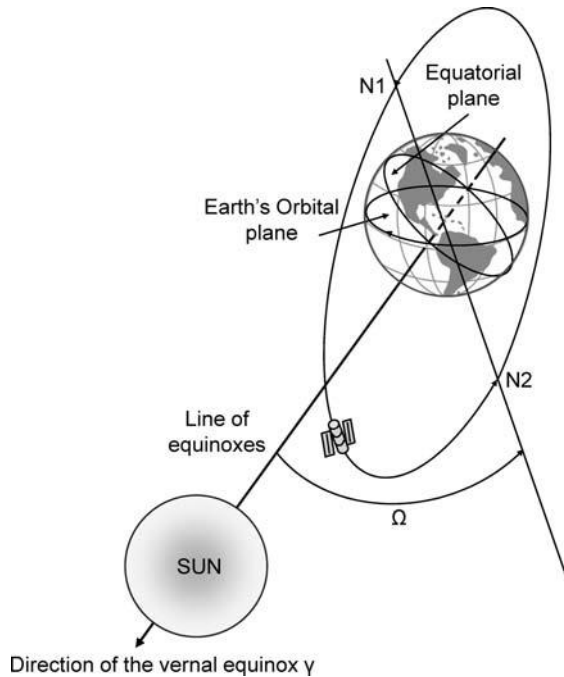


Figure 2.11 Vernal equinox

Solstices. Solstices are the times when the inclination angle is at its maximum, i.e. 23.4° . These also occur twice during a year on 20-21 June, called the summer solstice, and 21-22 December, called the winter solstice.

Apogee. Apogee is the point on the satellite orbit that is at the farthest distance from the centre of the Earth (Figure 2.12). The apogee distance can be computed from the known values of the orbit eccentricity e and the semi-major axis a from

$$\text{Apogee distance} = a(1 + e) \quad (2.20)$$

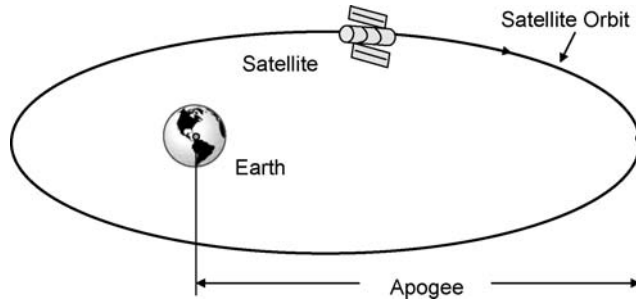


Figure 2.12 Apogee

The apogee distance can also be computed from the known values of the perigee distance and velocity at the perigee V_p from

$$V_p = \sqrt{\left(\frac{2\mu}{\text{Perigee distance}} - \frac{2\mu}{\text{Perigee distance} + \text{Apogee distance}} \right)} \quad (2.21)$$

where

$$V_p = V \left(\frac{d \cos \gamma}{\text{Perigee distance}} \right)$$

with V being the velocity of the satellite at a distance d from the centre of the Earth.

Perigee

. Perigee is the point on the orbit that is nearest to the centre of the Earth (Figure 2.13). The perigee distance can be computed from the known values of orbit

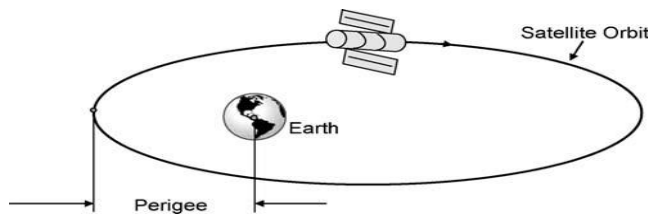


Figure 2.13 Perigee

eccentricity e and the semi-major axis a from

$$\text{Perigee distance} = a(1 - e) \quad (2.22)$$

Eccentricity. The orbit eccentricity e is the ratio of the distance between the centre of the ellipse and the centre of the Earth to the semi-major axis of the ellipse. It can be computed from any of the following expressions:

$$e = \frac{\text{apogee} - \text{perigee}}{\text{apogee} + \text{perigee}} \quad (2.23)$$

$$e = \frac{\text{apogee} - \text{perigee}}{2a} \quad (2.24)$$

Thus $e = \sqrt{(a^2 - b^2)/a^2}$, where a and b are semi-major and semi-minor axes respectively.

Semi-major axis. This is a geometrical parameter of an elliptical orbit. It can, however, be computed from known values of apogee and perigee distances as

$$a = \frac{\text{apogee} + \text{perigee}}{2} \quad (2.25)$$

Right ascension of the ascending node. The right ascension of the ascending node is the orientation of the line of nodes, which is the line joining the ascending and descending nodes, with respect to the direction of the vernal equinox.

It is expressed as an angle Ω measured from the vernal equinox towards the line of nodes in the direction of rotation of Earth (Figure 2.14). The angle could be anywhere from 0° to 360° .

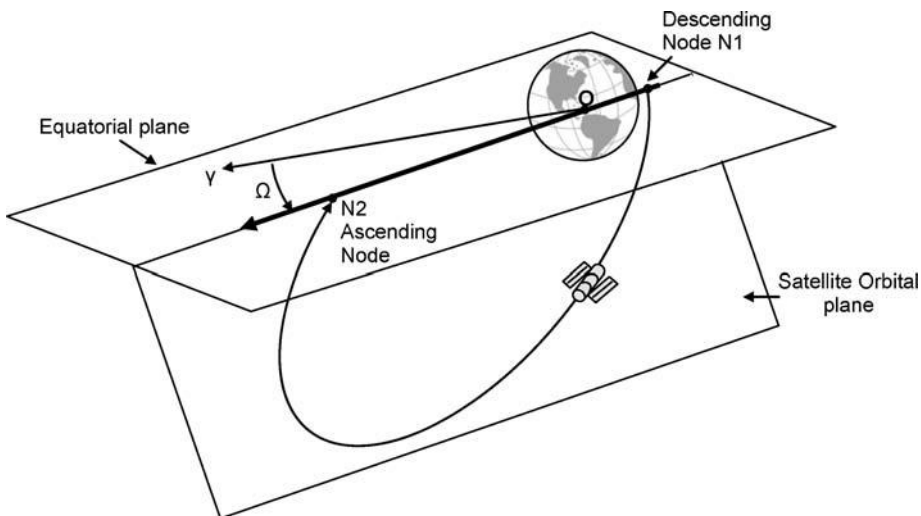


Figure 2.14 Right ascension of the ascending node

Angle Ω can be computed as the difference between two angles. One is the angle α between the direction of the vernal equinox and the longitude of the injection point and the other is the angle β between the line of nodes and the longitude of the injection point, as shown in Figure 2.15. Angle β can be computed from

$$\sin \beta = \frac{\cos i \sin l}{\cos l \sin i} \quad (2.26)$$

where $\angle i$ is the orbit inclination and l is the latitude at the injection point.

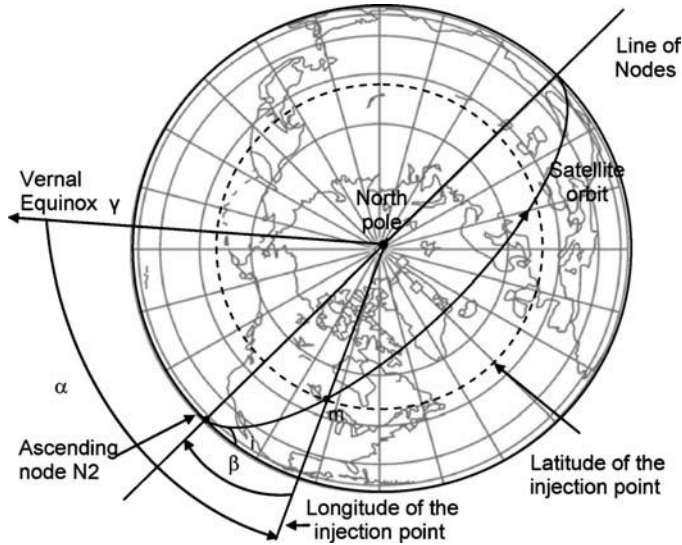


Figure 2.15 Computation of the right ascension of the ascending node

Inclination. Inclination is the angle that the orbital plane of the satellite makes with the Earth's equatorial plane.

It is measured as follows.

The line of nodes divides both the Earth's equatorial plane and the satellite's orbital plane into two halves. The **inclination** of the satellite's orbit is defined as the angle between two specific halves: the half of the satellite's orbital plane that contains the satellite's trajectory from the descending node to the ascending node, and the corresponding half of the Earth's equatorial plane that contains the trajectory of a point on the equator moving from point n_1 to point n_2 . Here, n_1 and n_2 are the points on the Earth's surface located directly beneath the descending and ascending nodes, respectively. (Figure 2.16).

The inclination angle can be determined from the latitude l at the injection point and the angle A_z between the projection of the satellite's velocity vector on the local horizontal and north. It is given by

$$\cos i = \sin A_z \cos l \quad (2.27)$$

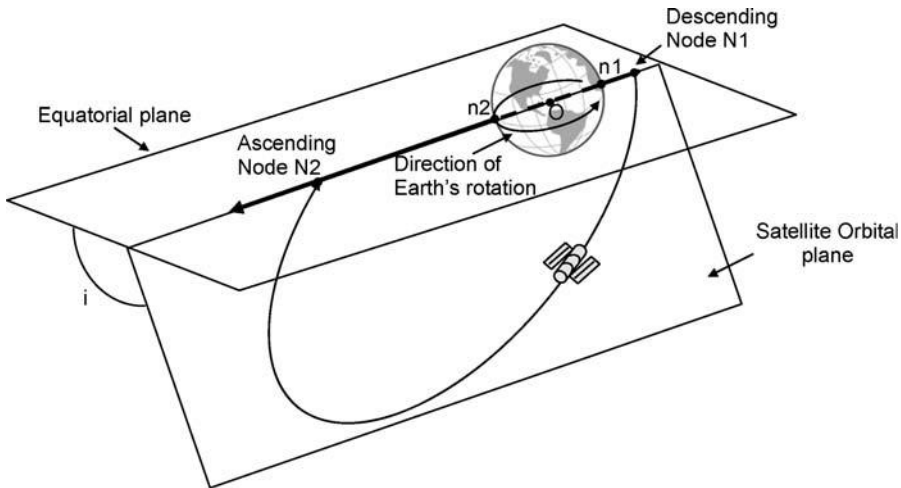


Figure 2.16 Angle of inclination

Argument of the perigee. This parameter defines the location of the major axis of the satellite orbit. It is measured as the angle ω between the line joining the perigee and the centre of the Earth and the line of nodes from the ascending node to the descending node in the same direction as that of the satellite orbit (Figure 2.17).

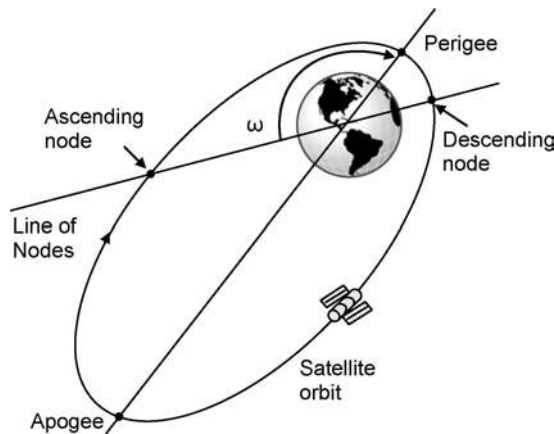


Figure 2.17 Argument of perigee

True anomaly of the satellite. This parameter is used to indicate the position of the satellite in its orbit. This is done by defining an angle ϑ , called the true anomaly of the satellite, formed by the line joining the perigee and the centre of the Earth with the line joining the satellite and the centre of the Earth (Figure 2.18).

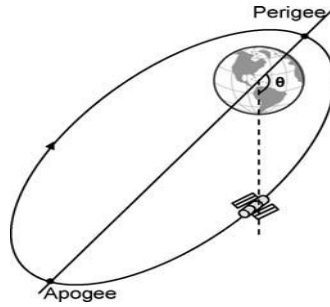


Figure 2.18 True anomaly of a satellite

Angles defining the direction of the satellite. The direction of the satellite is defined by two angles, the first by angle γ between the direction of the satellite's velocity vector and its projection in the local horizontal and the second by angle A_z between the north and the projection of the satellite's velocity vector on the local horizontal (Figure 2.19).

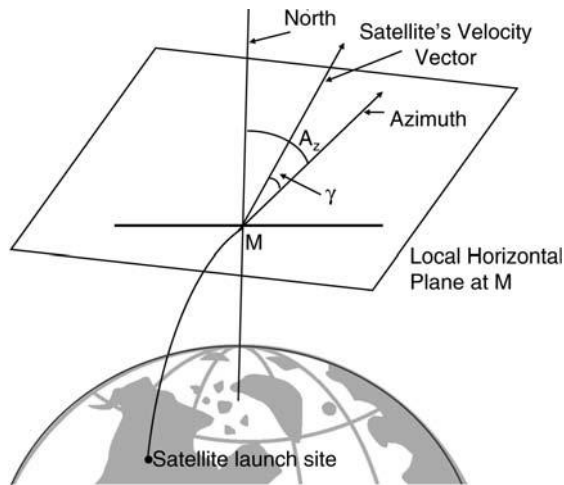


Figure 2.19 Angles defining the direction of the satellite

Injection Velocity and Resulting Satellite Trajectories

The horizontal velocity with which a satellite is injected into space by the launch vehicle with the intention of imparting a specific trajectory to the satellite has a direct bearing on the satellite trajectory. The phenomenon is best explained in terms of the three cosmic velocities. The general expression for the velocity of a satellite at the perigee point (V_P), assuming an elliptical orbit, is given by

$$V_P = \sqrt{\left[\left(\frac{2\mu}{r} \right) - \left(\frac{2\mu}{R+r} \right) \right]} \quad (2.28)$$

where

R = apogee distance

r = perigee distance

$\mu = GM = \text{constant}$

The first cosmic velocity V_1 is the one at which apogee and perigee distances are equal, i.e. $R = r$, and the orbit is circular. The above expression then reduces to

$$V_1 = \sqrt{\left(\frac{\mu}{r} \right)} \quad (2.29)$$

Thus, irrespective of the distance r of the satellite from the centre of the Earth, if the injection velocity is equal to the first cosmic velocity, also sometimes called the first orbital velocity, the satellite follows a circular orbit (Figure 2.27) and moves with a uniform velocity equal to $\sqrt{(\mu/r)}$.

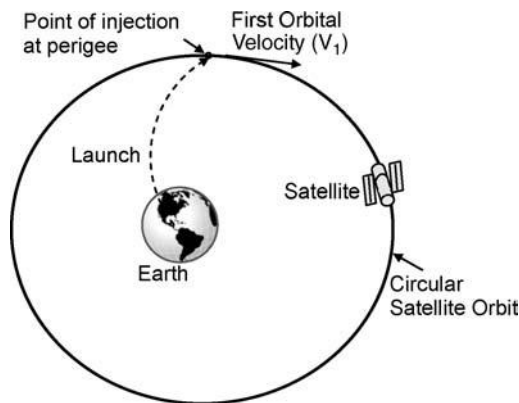


Figure 2.27 Satellite's path where the injection velocity is equal to the first orbital velocity

If the injection velocity happens to be less than the first cosmic velocity, the satellite follows a ballistic trajectory and falls back to Earth. In this case, the orbit is elliptical and the injection point is at the apogee and not the perigee. If the perigee lies in the atmosphere or exists only virtually below the surface of the Earth, the satellite accomplishes a ballistic flight and falls back to Earth (Figure 2.28).

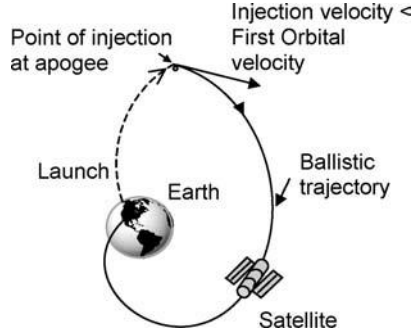


Figure 2.28 Satellite's path where the injection velocity is less than the first orbital velocity

For injection velocity greater than the first cosmic velocity and less than the second cosmic velocity, i.e. $V > \sqrt{\mu/r}$ and $V < \sqrt{2\mu/r}$, the orbit is elliptical and eccentric. The orbit eccentricity is between 0 and 1. The injection point in this case is the perigee and the apogee distance attained in the resultant elliptical orbit depends upon the injection velocity.

The higher the injection velocity, the greater is the apogee distance. The apogee distance can also be computed from the known value of injection velocity, which is also the velocity at the perigee point as the perigee coincides with the injection point, and the velocity v at any other point in the orbit distant d from the centre of the Earth using

$$V_p = \sqrt{\left[(2\mu/r) - \left(\frac{2\mu}{R+r} \right) \right]} = \frac{vd \cos \gamma}{r} \quad (2.30)$$

Then the injection velocity equals $\sqrt{2\mu/r}$, the apogee distance R becomes infinite and the orbit takes the shape of a parabola (Figure 2.29) and the orbit eccentricity is 1. This is the second cosmic velocity v_2 . At this velocity, the satellite escapes Earth's gravitational pull. For an injection velocity greater than the second cosmic velocity, the trajectory is hyperbolic within the solar system and the orbit eccentricity is greater than 1.

If the injection velocity is increased further, a stage is reached where the satellite succeeds in escaping from the solar system. This is known as the third cosmic velocity and is related to the motion of planet Earth around the sun. The third cosmic velocity (V_3) is mathematically expressed as

$$V_3 = \sqrt{\left[\frac{2\mu}{r} - V_t^2(3 - 2\sqrt{2}) \right]} \quad (2.31)$$

where V_t is the speed of Earth's revolution around the sun.

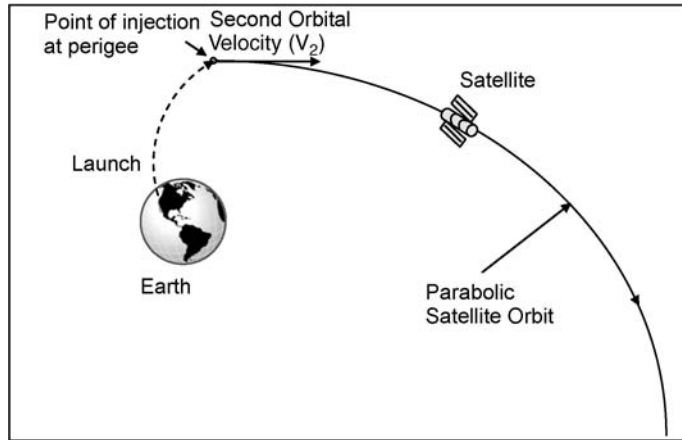


Figure 2.29 Satellite's path where the injection velocity is equal to the second orbital velocity

Beyond the third cosmic velocity, there is a region of hyperbolic flights outside the solar system. Coming back to elliptical orbits, the greater the injection velocity from the first cosmic velocity, the greater is the apogee distance. This is evident from the generalized expression for the velocity of the satellite in elliptical orbits according to which

$$V_p = \text{velocity at the perigee point}$$

$$= \sqrt{\left[(2\mu) \left(\frac{1}{r} - \frac{1}{R+r} \right) \right]}$$

Types of Satellite Orbits

The satellite orbits can be classified on the basis of:

1. Orientation of the orbital plane
2. Eccentricity
3. Distance from Earth

Orientation of the Orbital Plane

The orbits can be classified as equatorial orbits, polar orbits and inclined orbits. In the case of an equatorial orbit, the angle of inclination is zero, i.e. the orbital plane of the satellite coincides with the Earth's equatorial plane (Figure 2.32).

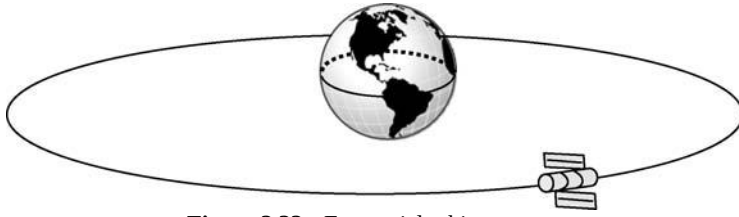


Figure 2.32 Equatorial orbit

A satellite in the equatorial orbit has a latitude of 0° . For an angle of inclination equal to 90° , the satellite is said to be in the polar orbit (Figure 2.33). For an angle of inclination between 0° and 180° , the orbit is said to be an inclined orbit.



Figure 2.33 Polar orbit

For inclinations between 0° and 90° , the satellite travels in the same direction as the direction of rotation of the Earth. The orbit in this case is referred to as a direct or prograde orbit (Figure 2.34).



Figure 2.34 Prograde orbit

For inclinations between 90° and 180° , the satellite orbits in a direction opposite to the direction of rotation of the Earth and the orbit in this case is called a retrograde orbit (Figure 2.35).



Figure 2.35 Retrograde orbit

Eccentricity of the Orbit

On the basis of eccentricity, the orbits are classified as elliptical (Figure 2.36 (a)) and circular (Figure 2.36 (b)) orbits. Needless to say, when the orbit eccentricity lies between 0 and 1, the orbit is elliptical with the centre of the Earth lying at one of the foci of the ellipse.

When the eccentricity is zero, the orbit becomes circular. It may be mentioned here that all circular orbits are eccentric to some extent. As an example, the eccentricity of orbit of geostationary

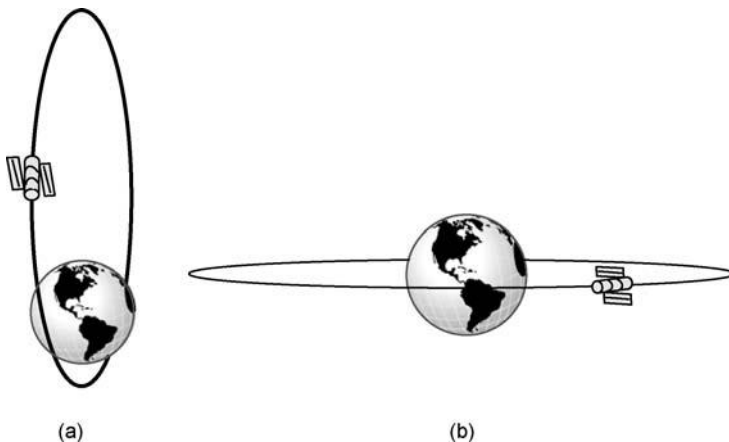


Figure 2.36 (a) Elliptical orbit and (b) circular orbit

Molniya Orbit

Eccentric, inclined and elliptical orbits are used to cover higher latitudes, which are otherwise not covered by geostationary orbits. A practical example of this type of orbit is the Molniya orbit (Figure 2.37).

The Molniya orbit serves the purpose of a geosynchronous orbit for high latitude regions. It is a 12 hour orbit and a satellite in this orbit spends about 8 hours above a particular high latitude station before diving down to a low level perigee at an equally high southern latitude. Usually, three satellites at different phases of the same Molniya orbit are capable of providing an uninterrupted service.

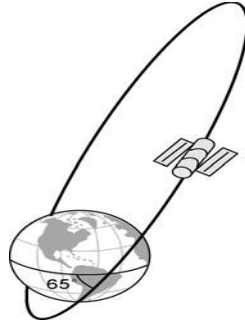


Figure 2.37 Molniya orbit

Distance from Earth

Depending upon the intended mission, satellites may be placed in orbits at varying distances from the surface of the Earth. Depending upon the distance, these are classified as low Earth orbits (LEOs), medium Earth orbits (MEOs) and geostationary Earth orbits (GEOs), as shown in Figure 2.38.

Satellites in the low Earth orbit (LEO) circle Earth at a height of around 160 to 500 km above the surface of the Earth.

These satellites, being closer to the surface of the Earth, have much shorter orbital periods and smaller signal propagation delays.

A lower propagation delay makes them highly suitable for communication applications. Due to lower propagation paths, the power required for signal transmission is also less, with the result that the satellites are of small physical size and are inexpensive to build.

However, due to a shorter orbital period, of the order of an hour and a half or so, these satellites remain over a particular ground station for a short time.

Hence, several of these satellites are needed for 24 hour coverage.

One important application of LEO satellites for communication is the project Iridium, which is a global communication system conceived by Motorola (Figure 2.39). A total of 66 satellites are arranged in a distributed architecture, with each satellite carrying 1/66 of the total system capacity. The system is intended to provide a variety of telecommunication services at the global level.

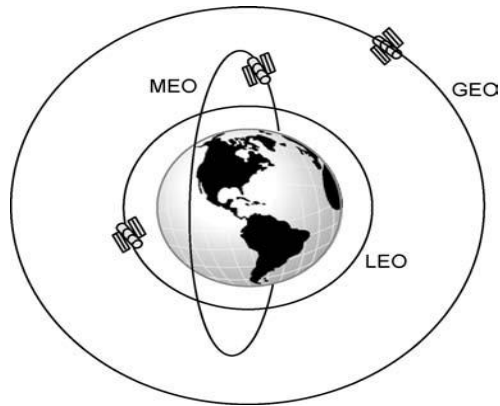


Figure 2.38 LEO, MEO and GEO



Figure 2.39 Iridium constellation of satellites

Medium Earth orbit (MEO) satellites orbit at a distance of approximately 10 000 to 20 000 km above the surface of the Earth.

They have an orbital period of 6 to 12 hours.

These satellites stay in sight over a particular region of Earth for a longer time. The transmission distance and propagation delays are greater than those for LEO satellites.

These orbits are generally polar in nature and are mainly used for communication and navigation applications.

A geosynchronous Earth orbit is a prograde orbit whose orbital period is equal to Earth's rotational period. If such an orbit were in the plane of the equator and circular, it would remain stationary with respect to a given point on the Earth. These orbits are referred to as the geostationary Earth orbits (GEOs).

For the satellite to have such an orbital velocity, it needs to be at a height of about 36 000 km, 35 786 km to be precise, above the surface of the Earth.

To be more precise and technical, in order to remain above the same point on the Earth's surface, a satellite must fulfil the following conditions:

1. It must have a constant latitude, which is possible only at 0° latitude.
2. The orbit inclination should be zero.
3. It should have a constant longitude and thus have a uniform angular velocity, which is possible when the orbit is circular.
4. The orbital period should be equal to 23 hours 56 minutes, which implies that the satellite must orbit at a height of 35 786 km above the surface of the Earth.
5. The satellite motion must be from west to east.

In the case where these conditions are fulfilled, then as the satellite moves from a position O_1 to O_2 in its orbit, a point vertically below on the equator moves with the same angular velocity and moves from E_1 to E_2 , as shown in Figure 2.40.

Satellites in geostationary orbits play an essential role in relaying communication and TV broadcast signals around the globe. They also perform meteorological and military surveillance functions very effectively.

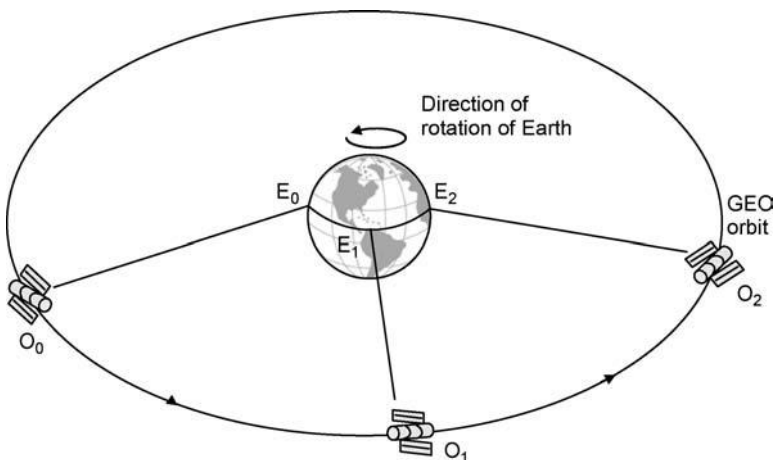


Figure 2.40 GEO satellites appear stationary wrt to a point on Earth

Sun-synchronous Orbit

A sun-synchronous orbit, also known as a helio-synchronous orbit, is one that lies in a plane that maintains a fixed angle with respect to the Earth–sun direction. The orbital plane has a fixed orientation with respect to the Earth–sun direction and the angle between the orbital plane and the Earth–sun line remains constant throughout the year, as shown in Figure 2.41. Satellites in sun-synchronous orbits are particularly suited to applications like passive remote sensing, meteorological, military reconnaissance and atmospheric studies.

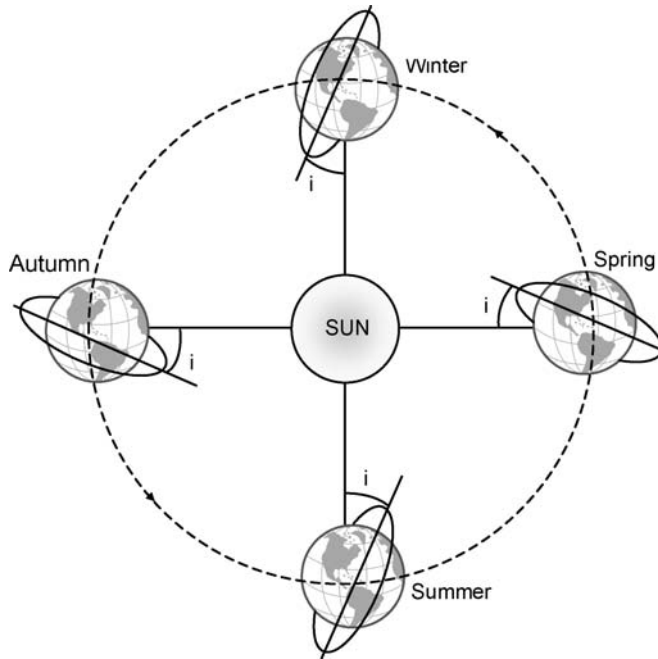


Figure 2.41 Sun-synchronous orbit

As a result of this property, sun-synchronous orbits ensure that:

1. The satellite passes over a given location on Earth every time at the same local solar time, thereby guaranteeing almost the same illumination conditions, varying only with seasons.
2. The satellite ensures coverage of the whole surface of the Earth, being quasi-polar in nature.

Every time a sun-synchronous satellite completes one revolution around Earth, it traverses a thin strip on the surface of the Earth. During the next revolution it traverses another strip shifted westwards and the process of shift continues with successive revolutions, as shown in Figure 2.42.

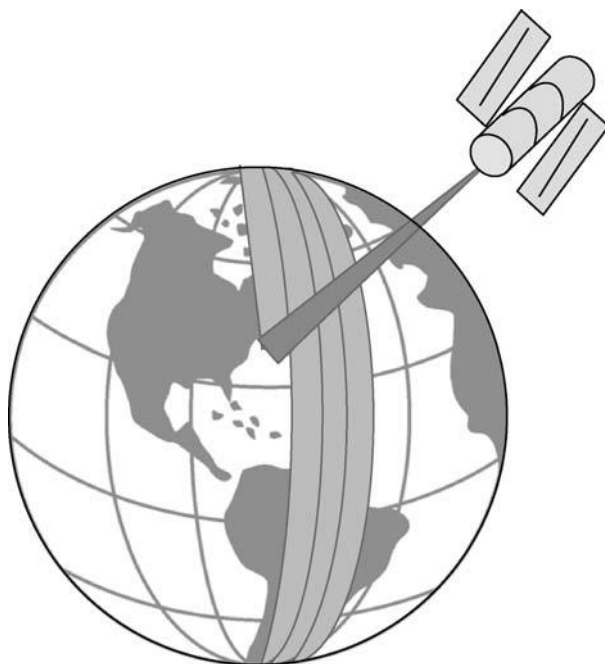


Figure 2.42 Earth coverage of sun-synchronous satellites

Depending upon the orbital parameters and speed of rotation of Earth, after making a certain number of revolutions around Earth, it comes back close to the first strip that it had traversed. It may not exactly overlap the first strip.

The number of revolutions required before the satellite repeats the same strip is called one complete orbital cycle, which is basically the time that elapses before the satellite revisits a given location in the same direction.

Numericals

Problem 2.1 A satellite is orbiting Earth in a uniform circular orbit at a height of 630 km from the surface of Earth. Assuming the radius of Earth and its mass to be 6370 km and 5.98×10^{24} kg respectively, determine the velocity of the satellite (Take the gravitational constant $G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$)

Solution:

Orbit radius $R = 6370 + 630 = 7000 \text{ km} = 7\,000\,000 \text{ m}$

$$\begin{aligned}\text{Also, constant } \mu &= GM = 6.67 \times 10^{-11} \times 5.98 \times 10^{24} \\ &= 39.8 \times 10^{13} \text{ N m}^2/\text{kg} \\ &= 39.8 \times 10^{13} \text{ m}^3/\text{s}^2\end{aligned}$$

The velocity of the satellite can be computed from

$$V = \sqrt{\left(\frac{\mu}{R}\right)} = \sqrt{\left(\frac{39.8 \times 10^{13}}{7\,000\,000}\right)} = 7.54 \text{ km/s}$$

Problem 2.2

The apogee and perigee distances of a satellite orbiting in an elliptical orbit are respectively 45 000 km and 7000 km. Determine the following:

1. Semi-major axis of the elliptical orbit
2. Orbit eccentricity
3. Distance between the centre of the Earth and the centre of the elliptical orbit

Solution:

$$\begin{aligned}\text{1. Semi-major axis of the elliptical orbit } a &= \frac{\text{apogee} + \text{perigee}}{2} \\ &= \frac{45\,000 + 7000}{2} = 26\,000 \text{ km}\end{aligned}$$

$$\text{2. Eccentricity } e = \frac{\text{apogee} - \text{perigee}}{2a} = \frac{45\,000 - 7000}{2 \times 26\,000} = \frac{38\,000}{52\,000} = 0.73$$

$$\begin{aligned}\text{3. Distance between the centre of the Earth and the centre of the ellipse} &= ae \\ &= 26\,000 \times 0.73 \\ &= 18\,980 \text{ km}\end{aligned}$$

Problem 2.3

A satellite is moving in an elliptical orbit with the major axis equal to 42 000 km. If the perigee distance is 8000 km, find the apogee and the orbit eccentricity.

Solution:

Major axis = 42 000 km

Also, major axis = apogee + perigee = 42 000 km

Therefore apogee = 42 000 – 8000 = 34 000 km

$$\begin{aligned}\text{Also, eccentricity } e &= \frac{\text{apogee} - \text{perigee}}{\text{major axis}} \\ &= \frac{34\,000 - 8000}{42\,000} \\ &= \frac{26\,000}{42\,000} = 0.62\end{aligned}$$

Problem 2.4

Refer to the satellite orbit of Figure 2.20. Determine the apogee and perigee distances if the orbit eccentricity is 0.6.

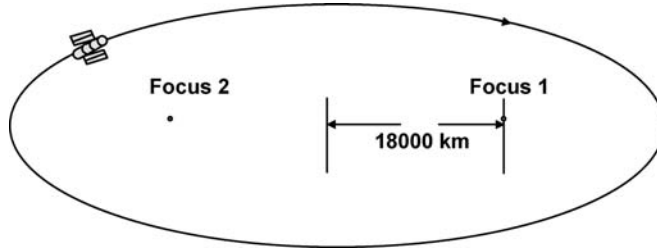


Figure 2.20 Figure for Problem 2.4

Solution: If e is the orbit eccentricity and a the semi-major axis of the elliptical orbit, then the distance between the centre of the Earth and the centre of the ellipse is equal to ae . Therefore $ae = 18\,000\text{ km}$

This gives $a = 18\,000/e = 18\,000/0.6 = 30\,000\text{ km}$

Apogee distance $= a(1 + e) = 30\,000 \times (1 + 0.6) = 48\,000\text{ km}$

Perigee distance $= a(1 - e) = 30\,000 \times (1 - 0.6) = 12\,000\text{ km}$

Problem 2.5

The difference between the furthest and the closest points in a satellite's elliptical orbit from the surface of the Earth is $30\,000\text{ km}$ and the sum of the distances is $50\,000\text{ km}$. If the mean radius of the Earth is considered to be 6400 km , determine orbit eccentricity.

Solution: Apogee – Perigee $= 30\,000\text{ km}$ as the radius of the Earth will cancel in this case

$$\text{Apogee} + \text{Perigee} = 50\,000 + 2 \times 6400 = 62\,800\text{ km}$$

$$\text{Orbit eccentricity} = (\text{Apogee} - \text{Perigee})/(\text{Apogee} + \text{Perigee}) = 30\,000/62\,800 = 0.478$$

Problem 2.6

Refer to Figure 2.21. Satellite A is orbiting Earth in a near-Earth circular orbit of radius 7000 km . Satellite B is orbiting Earth in an elliptical orbit with apogee and perigee distances of $47\,000\text{ km}$ and 7000 km respectively. Determine the velocities of the two satellites at point X. (Take $\mu = 39.8 \times 10^{13}\text{ m}^3/\text{s}^2$.)

Solution: The velocity of a satellite moving in a circular orbit is constant throughout the orbit and is given by

$$V = \sqrt{\left(\frac{\mu}{R}\right)}$$

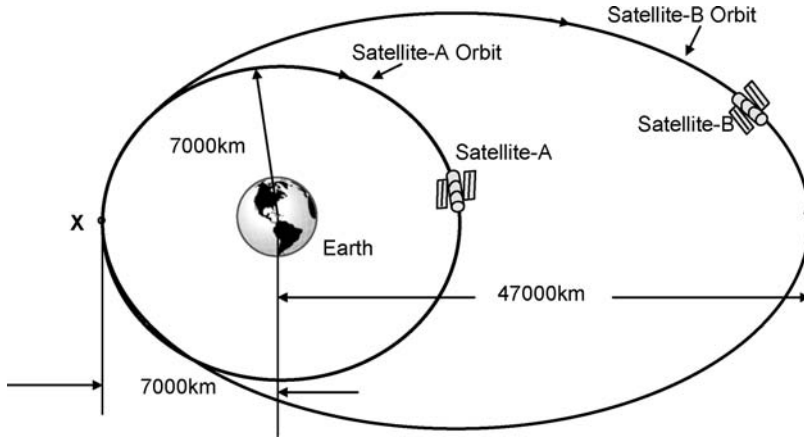


Figure 2.21 Figure for Problem 2.6

Therefore the velocity of satellite A at point X = $\sqrt{\left(\frac{39.8 \times 10^{13}}{7000000}\right)}$

$$= 7.54 \text{ km/s}$$

The velocity of the satellite at any point in an elliptical orbit is given by

$$v = \sqrt{\left[\mu \left(\frac{2}{R} - \frac{1}{a}\right)\right]}$$

where a is the semi-major axis and R is the distance of the point in question from the centre of the Earth. Here $R = 7000 \text{ km}$ and $a = (47000 + 7000)/2 = 27000 \text{ km}$. Therefore, velocity of satellite B at point X is given by

$$v = \sqrt{\left[(39.8 \times 10^{13}) \times \left(\frac{2}{7000000} - \frac{1}{27000000}\right)\right]} = 9.946 \text{ km/s}$$

Problem 2.7

Refer to Figure 2.22. Satellite A is orbiting Earth in an equatorial circular orbit of radius 42 000 km. Satellite B is orbiting Earth in an elliptical orbit with apogee and perigee distances of 42 000 km and 7000 km respectively. Determine the velocities of the two satellites at point X. (Take $\mu = 39.8 \times 10^{13} \text{ m}^3/\text{s}^2$.)

Solution: The velocity of a satellite moving in a circular orbit is constant throughout the orbit and is given by

$$v = \sqrt{\left(\frac{\mu}{R}\right)}$$

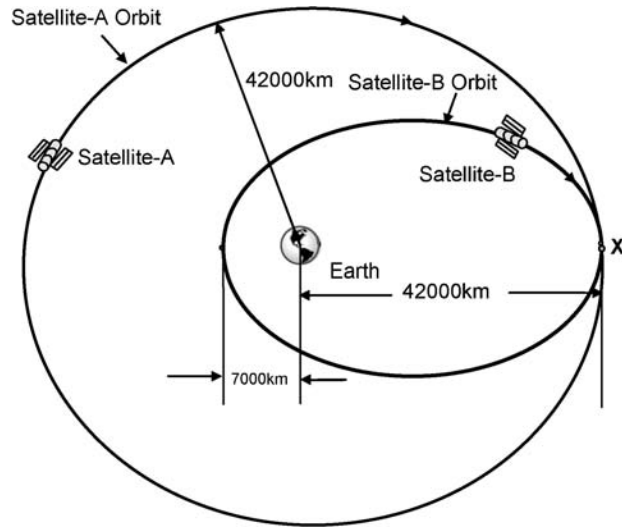


Figure 2.22 Figure for Problem 2.7

Therefore, the velocity of satellite A at point X =
$$\sqrt{\left(\frac{39.8 \times 10^{13}}{42\,000\,000}\right)}$$

$$= 3.078 \text{ km/s}$$

The velocity of the satellite at any point in an elliptical orbit is given by

$$v = \sqrt{\left[\mu \left(\frac{2}{R} - \frac{1}{a}\right)\right]}$$

where a is the semi-major axis and R is the distance of the point in question from the centre of the Earth. Here $R = 42\,000 \text{ km}$ and $a = (42\,000 + 7\,000)/2 = 24\,500 \text{ km}$. Therefore, velocity of satellite B at point X is given by

$$v = \sqrt{\left[(39.8 \times 10^{13}) \times \left(\frac{2}{42\,000\,000} - \frac{1}{24\,500\,000}\right)\right]} = 1.645 \text{ km/s}$$

Problem 2.8

Refer to Figure 2.23. Satellite A is orbiting Earth in a circular orbit of radius 25 000 km. Satellite B is orbiting Earth in an elliptical orbit with apogee and perigee distances of 43 000 km and 7 000 km respectively. Determine the velocities of the two satellites at the indicated points X and Y. (Take $\mu = 39.8 \times 10^{13} \text{ m}^3/\text{s}^2$.)

Solution: The velocity of a satellite moving in a circular orbit is constant throughout the orbit and is given by

$$v = \sqrt{\left(\frac{\mu}{R}\right)}$$

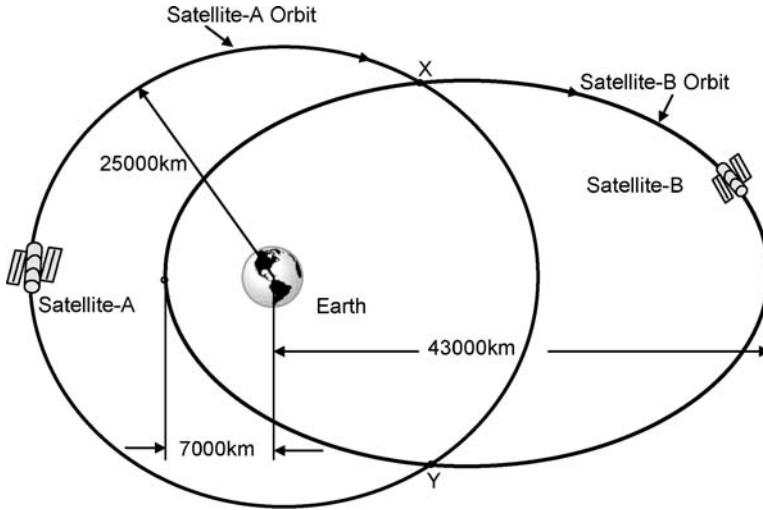


Figure 2.23 Figure for Problem 2.8

Therefore the velocity of satellite A at points X and Y = $\sqrt{\left(\frac{39.8 \times 10^{13}}{25\,000\,000}\right)} = 3.989 \text{ km/s}$
 The velocity of the satellite at any point in an elliptical orbit is given by

$$v = \sqrt{\left[\mu \left(\frac{2}{R} - \frac{1}{a}\right)\right]}$$

where a is the semi-major axis and R is the distance of the point in question from the centre of the Earth. For satellite B, at points X and Y, $R = 25\,000 \text{ km}$ and $a = (43\,000 + 7\,000)/2 = 25\,000 \text{ km}$. Therefore, velocity of satellite B at points X and Y

$$v = \sqrt{\left[(39.8 \times 10^{13}) \times \left(\frac{2}{25\,000\,000} - \frac{1}{25\,000\,000}\right)\right]} = 3.989 \text{ m/s}$$

Please note: In Problems 2.6, 2.7 and 2.8, three different cases of satellites moving in circular and elliptical orbits have been considered. Note that the velocity of the satellite at a point in the elliptical orbit is more than the velocity of the satellite in a circular orbit if the distance of the satellite from the centre of the Earth happens to be less than the radius of the circular orbit. It will be less than the velocity in the circular orbit if the distance of the satellite in the elliptical orbit happens to be more than the radius of the circular orbit. The two velocities are the same at a point where the distance of the satellite from the centre of the Earth in the elliptical orbit equals the radius of the circular orbit provided that the semi-major axis of the ellipse also equals the radius of the circular orbit.

Problem 2.9

Calculate the orbital period of a satellite moving in an elliptical orbit having a major axis of 50 000 km. (Take $\mu = 39.8 \times 10^{13} \text{ Nm}^2/\text{kg}$).

Solution: Semi-major axis, $a = 50\,000/2 = 25\,000 \text{ km}$

$$\begin{aligned} \text{Orbital time period, } T &= 2\pi \times \sqrt{(a^3/\mu)} = 2 \times 3.14 \times \sqrt{(25\,000\,000)^3/(39.8 \times 10^{13})} \\ &= 39\,344 \text{ seconds} = 10 \text{ hours } 55 \text{ minutes } 44 \text{ seconds} \end{aligned}$$

Problem 2.10

Refer to Figure 2.24. The semi-major axes of the two satellites shown in the figure are 18 000 km (satellite 1) and 24 000 km (satellite 2). Determine the relationship between their orbital periods.

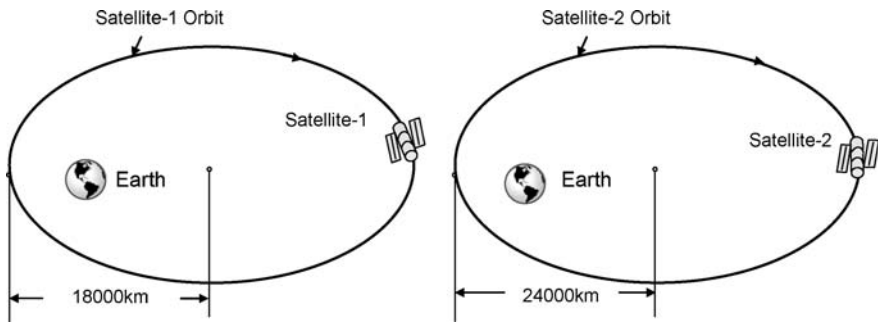


Figure 2.24 Figure for Problem 2.10

Solution: The orbital time period $T = 2\pi \times \sqrt{(a^3/\mu)}$ where a is the semi-major axis. If a_1 and a_2 are the semi-major axes in the two cases and T_1 and T_2 are their time periods, then

$$T_1 = 2\pi \times \sqrt{\left(\frac{a_1^3}{\mu}\right)}$$

and

$$T_2 = 2\pi \times \sqrt{\left(\frac{a_2^3}{\mu}\right)}$$

This gives

$$\frac{T_2}{T_1} = \left(\frac{a_2}{a_1}\right)^{3/2} = \left(\frac{24\,000}{18\,000}\right)^{3/2} = 1.54$$

The orbital period of satellite 2 is 1.54 times that of satellite 1.

Problem 2.11

The elliptical eccentric orbit of a satellite has its semi-major and semi-minor axes as 25 000 km and 18 330 km respectively. Determine the apogee and perigee distances.

Solution: If r_a and r_p are the apogee and perigee distances respectively, a and b are the semi-major and semi-minor axis respectively and e is the eccentricity, then

$$a = \frac{r_a + r_p}{2}$$

and

$$b = \sqrt{r_a r_p}$$

Now $(r_a + r_p)/2 = 25\,000$, which gives $r_a + r_p = 50\,000$

and $\sqrt{r_a r_p} = 18\,330$, which gives $r_a r_p = 335\,988\,900$

Substituting the value of r_p from these equations gives

$$r_a(50\,000 - r_a) = 335\,988\,900$$

which gives

$$r_a^2 - 50\,000r_a + 335\,988\,900 = 0$$

Solving this quadratic equation gives two solutions:

$$r_a = 42\,000 \text{ km}, 8000 \text{ km}$$

If the semi-major axis is 25 000 km, r_a cannot be 8000 km. Therefore

$$\text{Apogee distance } r_a = 42\,000 \text{ km}$$

and

$$\text{Perigee distance } r_p = 50\,000 - 42\,000 = 8000 \text{ km}$$

Alternative solution:

$$e = \frac{\sqrt{a^2 - b^2}}{a} = e = \frac{\sqrt{[(25\,000)^2 - (18\,330)^2]}}{25\,000} = 0.68$$

This gives

$$\text{Apogee distance} = a(1 + e) = 25\,000 \times 1.68 = 42\,000 \text{ km}$$

$$\text{and Perigee distance} = a(1 - e) = 25\,000 \times 0.32 = 8000 \text{ km}$$

Therefore, from both the methods,

$$\text{Apogee distance} = 42\,000 \text{ km and Perigee distance} = 8000 \text{ km}$$

Problem 2.12

Refer to Figure 2.25. The satellite is moving in an elliptical orbit with its semi-major and semi-minor axes as a and b respectively and an eccentricity of 0.6. The satellite takes

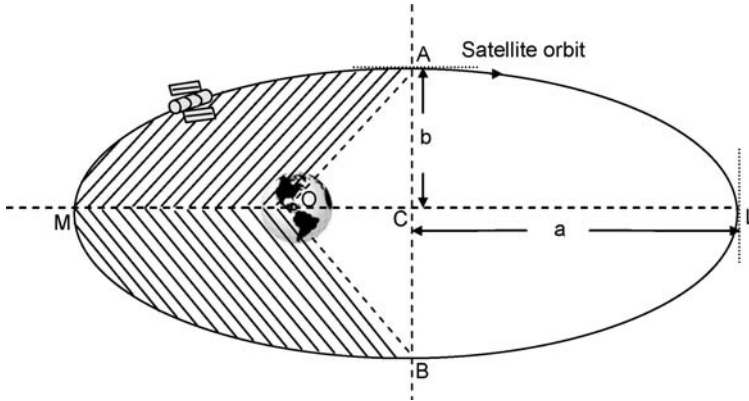


Figure 2.25 Figure for Problem 2.12

3 hours to move from point B to point A. How much time will it take to move from point A to point B?

Solution: This problem can be solved by applying Kepler's law of areas, which says that the line joining the satellite and centre of the Earth spans equal areas of the ellipse in equal times. Now, the area spanned when the satellite moves from point B to point A (area of the shaded region) is given by

$$\begin{aligned}
 \frac{1}{2} \times (\text{area of ellipse}) - (\text{area of } \triangle AOB) &= \frac{\pi ab}{2} - b \cdot OC \\
 &= \frac{\pi ab}{2} - bae \quad (\text{where } OC = ae) \\
 &= 1.57ab - 0.6ab \\
 &= 0.97ab
 \end{aligned}$$

The area spanned when the satellite moves from point A to point B (area of the unshaded region) is given by

$$\begin{aligned}
 \frac{1}{2} \times (\text{area of ellipse}) + (\text{area of } \triangle AOB) &= 1.57ab + 0.6ab \\
 &= 2.17ab
 \end{aligned}$$

The ratio of the two areas is given by $2.17/0.97 = 2.237$. The time taken by the satellite to move from point A to point B should therefore be 2.237 times the time taken by the satellite to move from point B to point A. Therefore

$$\text{Time taken} = 3 \times 2.237 = 6.711 \text{ hours}$$

Problem 2.13

The apogee and perigee distances of a certain elliptical satellite orbit are 42 000 km and 8000 km respectively. If the velocity at the perigee point is 9.142 km/s, what would be the velocity at the apogee point?

Solution: The product of the distance of a satellite from the centre of the Earth and the component of velocity vector in the direction of local horizontal at that point is constant at all possible points on the orbit; i.e. $dv \cos \gamma = \text{constant}$, where γ is the angle between the direction of motion of the satellite and the local horizontal. This has been explained earlier. The angle γ is zero at the apogee and perigee points. The expression thus reduces to $dv = \text{constant}$ for the apogee and perigee points. Therefore

$$\text{Velocity at the apogee} = \frac{8000 \times 9.142}{42\,000} = 1.741 \text{ km/s}$$

Problem 2.14

In the satellite orbit of Problem 2.13, at a certain point in the orbit at a distance of 16 000 km from the centre of the Earth, the direction of the satellite makes an angle of 56.245° with the local horizontal at that point (Figure 2.26). Determine the satellite velocity at that point.

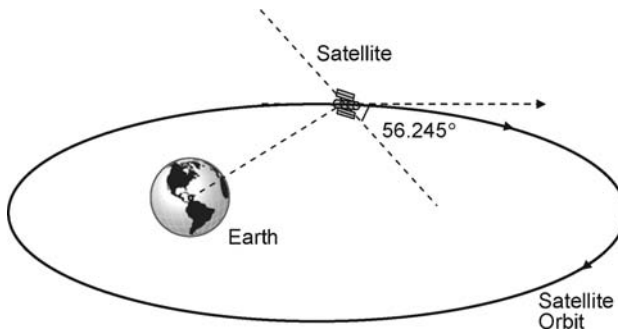


Figure 2.26 Figure for Problem 2.14

Solution: The product $dv \cos \gamma = 8000 \times 9.142$, which gives

$$\begin{aligned} v &= \frac{8000 \times 9.142}{16\,000 \times \cos 56.245^\circ} \\ &= \frac{73\,136}{16\,000 \times 0.555} = \frac{73\,136}{8880} = 8.236 \text{ km/s} \end{aligned}$$

Therefore

$$v = 8.236 \text{ km/s}$$

Problem 2.15

A satellite launched with an injection velocity of v_1 from a point above the surface of the Earth at a distance r from the centre of the Earth attains an elliptical orbit with an apogee distance of R_1 . The same satellite when launched with an injection velocity of v_2 from the same perigee distance attains an elliptical orbit with an apogee distance of R_2 . Derive the relationship between v_1 and v_2 in terms of r , R_1 and R_2 .

Solution: Refer to Figure 2.31. Expressions for the velocity at perigee points can be written as

$$v_1 = \sqrt{\left[(2\mu) \left(\frac{1}{r} - \frac{1}{R_1 + r} \right) \right]}$$

$$v_2 = \sqrt{\left[(2\mu) \left(\frac{1}{r} - \frac{1}{R_2 + r} \right) \right]}$$

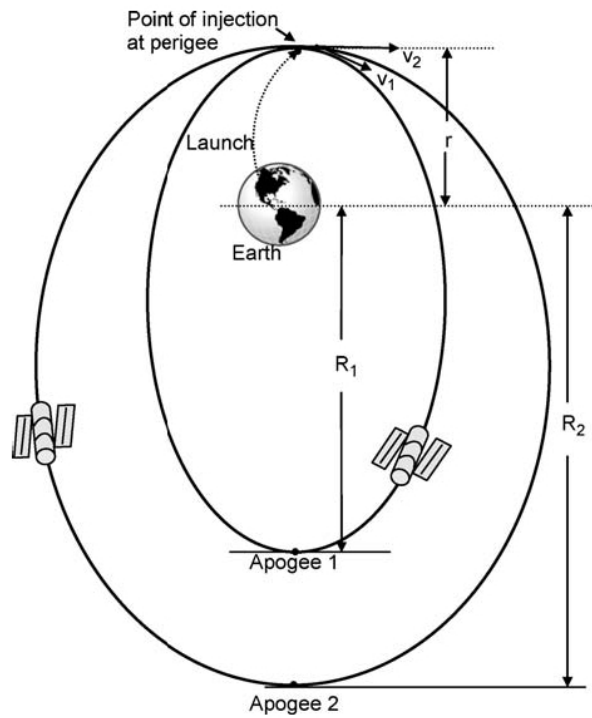


Figure 2.31 Figure for Problem 2.15

Squaring the two expressions and then taking the ratio of the two yields

$$\begin{aligned} \left(\frac{v_2}{v_1}\right)^2 &= \frac{(1/r) - 1/(R_2 + r)}{(1/r) - 1/(R_1 + r)} \\ &= \frac{R_2/r(R_2 + r)}{R_1/r(R_1 + r)} \\ &= \left(\frac{R_2}{R_1}\right) \left(\frac{R_1 + r}{R_2 + r}\right) \\ &= \frac{1 + r/R_1}{1 + r/R_2} \end{aligned}$$

Therefore,

$$\left(\frac{v_2}{v_1}\right)^2 = \frac{1 + r/R_1}{1 + r/R_2}$$

Problem 2.16

A satellite launched with an injection velocity of (v_1) from a point above the surface of the Earth at a distance of (r) of 8000 km from the centre of the Earth attains an elliptical orbit

with an apogee distance of 12 000 km. The same satellite when launched with an injection velocity of (v_2) that is 20 % higher than (v_1) from the same perigee distance attains an elliptical orbit with an apogee distance of (R_2). Determine the new apogee distance.

Solution:

We know that: $(v_2/v_1)^2 = [(1 + r/R_1)/(1 + r/R_2)]$

Substituting $v_2 = 1.2v_1$, we get $1.44 = [(1 + 8000/12\,000)/(1 + 8000/R_2)]$

or $1.44 \times 3 \times (1 + 8000/R_2) = 5$ or $(1 + 8000/R_2) = 1.1574$

$R_2 = 8000/0.1574 = 50\,826$ km

Problem 2.17

A rocket injects a satellite with a horizontal velocity of 8 km/s from a height of 1620 km from the surface of the Earth. What will be the velocity of the satellite at a point distant 10 000 km from the centre of the Earth, if the direction of the satellite makes an angle of 30° with the local horizontal at that point? Assume radius of the Earth to be 6380 km.

Solution:

We know that $V_2 = \sqrt{[2\mu/r] - [2\mu/(R+r)]} = (v.d.\cos\gamma)/r$

Here $V_2 = 8$ km/s, $r = 1620 + 6380 = 8000$ km, $d = 10\,000$ km and $\gamma = 30^\circ$

Therefore, $[v \times 10\,000 \times \cos 30^\circ / 8000] = 8$

or $v = (8 \times 8000) / (10\,000 \times 0.866) = 7.39$ km/s

Problem 2.18

A rocket injects a satellite with a certain horizontal velocity from a height of 620 km from the surface of the Earth. The velocity of the satellite at a point distant 9000 km from the centre of the Earth is observed to be 8 km/s. If the direction of the satellite makes an angle of 30° with the local horizontal at that point, determine the apogee distance of the satellite orbit. (Assume that the radius of the Earth is 6380 km and $\mu = 39.8 \times 10^{13} \text{ m}^3/\text{s}^2$.)

Solution:

It is known that $V_p = \sqrt{[\left(\frac{2\mu}{r}\right) - \left(\frac{2\mu}{R+r}\right)]} = \frac{vd\cos\gamma}{r}$

Here,

$r = 620 + 6380 = 7000$ km

$d = 9000$ km

$v = 8$ km/s

$\gamma = 30^\circ$

$R = \text{apogee distance} = ?$

Therefore,

$$\sqrt{[\left(\frac{2\mu}{r}\right) - \left(\frac{2\mu}{R+r}\right)]} = 8 \times 9000 \times \frac{0.866}{7000} = 8.90 \text{ km/s} = 8.90 \times 10^3 \text{ m/s}$$

or

$$\frac{2\mu}{r} - \frac{2\mu}{R+r} = (8.90 \times 10^3)^2 = 79.34 \times 10^6$$

or

$$\frac{2\mu}{R+r} = \frac{2 \times 39.8 \times 10^{13}}{7\,000\,000} - (79.34 \times 10^6) = 34.36 \times 10^6$$

This gives

$$R+r = \frac{2 \times 39.8 \times 10^{13}}{34.36 \times 10^6} = 2.317 \times 10^7 \text{ m}$$

$$\text{or } R = (2.317 \times 10^7 - 7\,000\,000) \text{ m} = 16\,170 \text{ km}$$

Problem 2.19

Verify that a geostationary satellite needs to be at a height of about 35 780 km above the surface of the Earth. Assume the radius of the Earth to be 6380 km and $\mu = 39.8 \times 10^{13} \text{ Nm}^2/\text{kg}$

Solution:

The orbital period of a satellite in circular orbit is given by $T = 2\pi r^{3/2} / \sqrt{\mu}$

Orbital period of a geostationary satellite is equal to 23 hours, 56 minutes, which is equal to 86 160 seconds.

Therefore, $2 \times 3.14 \times (r)^{3/2} / \sqrt{(39.8 \times 10^{13})} = 86\,160$

$6.28 \times (r)^{3/2} = 86\,160 \times \sqrt{(39.8 \times 10^{13})} = 1\,718\,886 \times 10^6$

or $(r)^{3/2} = 273\,708 \times 10^6$

or $r = (273\,708 \times 10^6)^{2/3} = 4215.5 \times 10^4 \text{ m} = 42\,155 \text{ km}$

Hence, height of satellite orbit above the surface of the Earth = $42\,155 - 6380 = 35\,775 \text{ km}$

Problem 2.20

A typical Molniya orbit has perigee and apogee heights above the surface of Earth as 400 km and 40 000 km respectively. Verify that the orbit has a 12 hour time period assuming the radius of the Earth to be 6380 km and $\mu = 39.8 \times 10^{13} \text{ N m}^2 / \text{kg}$.

Solution: The orbital period of a satellite in elliptical orbit is given by

$$T = \frac{2\pi a^{3/2}}{\sqrt{\mu}}$$

where a = semi-major axis of the elliptical orbit. Now $a = (40\,000 + 400 + 6380 + 6380)/2 = 26\,580 \text{ km}$. Therefore

$$\begin{aligned} T &= 6.28 \times \frac{(26\,580 \times 10^3)^{3/2}}{\sqrt{(39.8 \times 10^{13})}} \\ &= 43\,137 \text{ seconds} = 11.98 \text{ hours} \approx 12 \text{ hours} \end{aligned}$$

ORBITAL PERTURBATIONS

The satellite, once placed in its orbit, experiences various perturbing torques that cause variations in its orbital parameters with time.

These include

- Gravitational forces from other bodies like solar and lunar attraction
- Magnetic field interaction
- Solar radiation pressure
- Asymmetry of Earth's gravitational field etc.

Due to these factors, the satellite orbit tends to drift and its orientation also changes and hence the true orbit of the satellite is different from that defined using Kepler's laws. The satellite's position thus needs to be controlled both in the east–west as well as the north–south directions.

The east–west location needs to be maintained to prevent radio frequency (RF) interference from neighbouring satellites. It may be mentioned here that in the case of a geostationary satellite, a 1° drift in the east or west direction is equivalent to a drift of about 735 km along the orbit (Figure 3.27).

The north–south orientation has to be maintained to have proper satellite inclination.

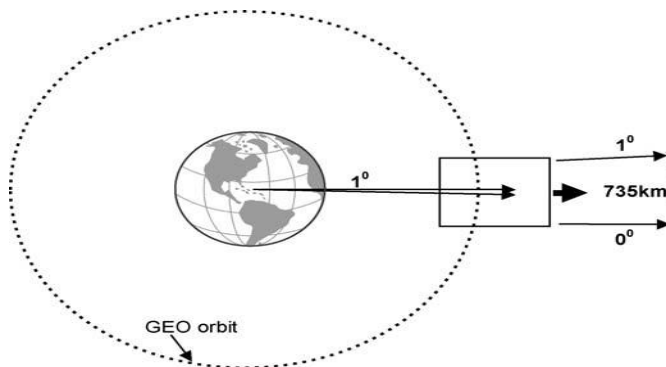


Figure 3.27 Drift of a geostationary satellite

The Earth is not a perfect sphere and is flattened at the poles. The equatorial diameter is about 20–40 km more than the average polar diameter.

The equatorial radius of the Earth is not constant.

The average density of Earth is not uniform.

All of this results in a non-uniform gravitational field around the Earth which in turn results in variation in gravitational force acting on the satellite due to the Earth.

The effect of variation in the gravitational field of the Earth on the satellite is more predominant for geostationary satellites than lower earth orbit satellites.

In the case of a geostationary satellite, these forces result in an acceleration or deceleration component that varies with the longitudinal location of the satellite.

In addition to the variation in the gravitational field of the Earth, the satellite is also subjected to the gravitational pulls of the sun and the moon.

The Earth's orbit around the sun is an ellipse whose plane is inclined at an angle of 7° with respect to the equatorial plane of the sun.

The Earth is tilted around 23° away from the normal to the ecliptic.

The moon revolves around the Earth with an inclination of around 5° to the equatorial plane of the Earth. Hence, the satellite in orbit is subjected to a variety of out-of-plane forces which change the inclination on the satellite's orbit.

The gravitational pulls of Earth, sun and moon have negligible effect of the satellites orbiting in LEO orbits, where the effect of atmospheric drag is more predominant. As the perturbed orbit is not an ellipse anymore, the satellite does not return to the same point in space after one revolution.

Anomalistic period :The time elapsed between the successive perigee passages is referred to as anomalistic period. The anomalistic period (T_A) is given by equation 3.17.

$$t_A = \frac{2\pi}{\omega_{\text{mod}}} \quad (3.17)$$

Where,

$$\omega_{\text{mod}} = \omega_0 \left[1 + \frac{K(1 - 1.5 \sin^2 i)}{a^2(1 - e^2)^{3/2}} \right]$$

ω_0 is the angular velocity for spherical Earth, $K = 66\,063.1704 \text{ km}^2$, a is the semi-major axis, e is the eccentricity and $i = \cos^{-1} W_Z$, W_Z is the Z axis component of the orbit normal.

The attitude and orbit control system maintains the satellite's position and its orientation and keeps the antenna pointed correctly in the desired direction

The orbit control is performed by

Thrusters and gas jets are used to correct the longitudinal drifts (in-plane changes) and the inclination changes (out-of- plane changes).

A different set of thrusters or gas jets is used for north-south and east-west manoeuvres.

3.3 Satellite Stabilization

Commonly employed techniques for satellite attitude control include:

1. Spin stabilization
2. Three-axis or body stabilization

SPIN STABILIZATION

Spin stabilization involves spinning the satellite around its axis (usually the axis perpendicular to the plane of its orbit). Once it's spinning, **the conservation of angular momentum** helps maintain the orientation of the satellite in space.

In a spin-stabilized satellite, the satellite body is spun at a rate between 30 and 100 rpm about an axis perpendicular to the orbital plane (Figure 3.28).

The rotating body offers inertial stiffness, which prevents the satellite from drifting from its desired orientation. Spin-stabilized satellites are generally cylindrical in shape. For stability, the satellite should be spun about its major axis, having a maximum moment of inertia. To maintain stability, the moment of inertia about the desired spin axis should at least be 10 % greater than the moment of inertia about the transverse axis.

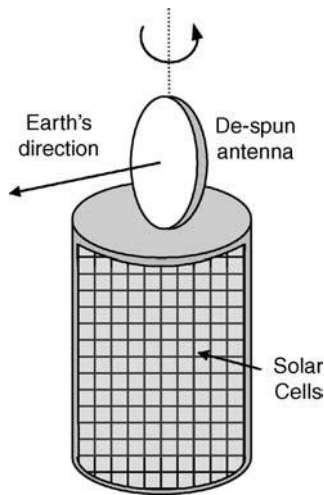


Figure 3.28 Spin stabilized satellite

There are two types of spinning configurations employed in spin-stabilized satellites.

- Simple spinner configuration
- Dual spinner configuration.

Simple spinner configuration: The satellite payload and other subsystems are placed in the spinning section, while the antenna and the feed are placed in the despun platform.

The despun platform is spun in a direction opposite to that of the spinning satellite.

Dual spinner configuration: the entire payload along with the antenna and the feed is placed on the de-spun platform and the other subsystems are located on the spinning body. Modern spin-stabilized satellites almost invariably employ the dual spinner configuration.

Three-axis or Body Stabilization

In the case of three-axis stabilization, also known as body stabilization, the stabilization is achieved by controlling the movement of the satellite along the three axes, i.e. yaw, pitch and roll, with respect to a reference (Figure 3.30).

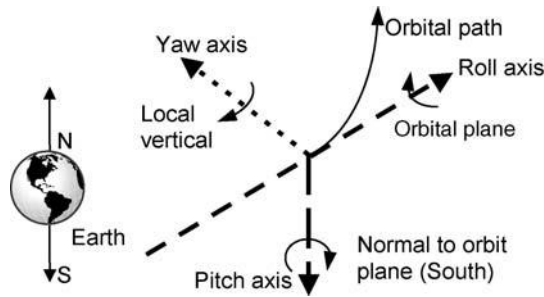


Figure 3.30 Three-axis stabilization

The system uses reaction wheels or momentum wheels to correct orbit perturbations. The stability of the three-axis system is provided by the active control system, which applies small corrective forces on the wheels to correct the undesirable changes in the satellite orbit.

The basic control technique used here is to speed up or slow down the momentum wheel depending upon the direction in which the satellite is perturbed. The satellite rotates in a direction opposite to that of speed change of the wheel.

For example, an increase in speed of the wheel in the clockwise direction will make the satellite to rotate in a counterclockwise direction. The momentum wheels rotate in one direction and can be twisted by a gimbal motor to provide the required dynamic force on the satellite.

An alternative approach is to use reaction wheels. Three reaction wheels are used, one for each axis. They can be rotated in either direction depending upon the active correction force. The satellite body is generally box shaped for three-axis stabilized satellites.

Comparison between Spin-stabilized and Three-axis Stabilized Satellites

1. In comparison to spin-stabilized satellites, three-axis stabilized satellites have more power generation capability and more additional mounting area available for complex antennae structures.
2. Spin-stabilized satellites are simpler in design and less expensive than three-axis stabilized satellites.
3. Three-axis stabilized satellites have the disadvantage that the extendible solar array used in these satellites are unable to provide power when the satellite is in the transfer orbit, as the array is still stored inside the satellite during this time.

Station Keeping

Station keeping is the process of maintenance of the satellite's orbit against different factors that cause temporal drift. Satellites need to have their orbits adjusted from time to time because the satellite, even though initially placed in the correct orbit, can undergo a progressive drift due to some natural forces such as minor gravitational perturbations due to the sun and moon, solar radiation pressure, Earth being an imperfect sphere, etc. The orbital adjustments are usually made by releasing jets of gas or by firing small rockets tied to the body of the satellite.

In the case of spin-stabilized satellites, station keeping in the north–south direction is maintained by firing thrusters parallel to the spin axis in a continuous mode. The east–west station keeping is obtained by firing thrusters perpendicular to the spin axis.

In the case of three-axis stabilization, station keeping is achieved by firing thrusters in the east–west or the north–south directions in a continuous mode

Orbital Effects on Satellite's Performance

The satellite is revolving constantly around the Earth. The motion of the satellite has significant effects on its performance. These include the Doppler shift, effect due to variation in the orbital distance, effect of solar eclipse and sun's transit outage.

Doppler Shift

The geostationary satellites appear stationary with respect to an Earth station terminal whereas in the case of satellites orbiting in low Earth orbits, the satellite is in relative motion with respect to the terminal. However, in the case of geostationary satellites also there are some variations between the satellite and the Earth station terminal. As the satellite is moving with respect to the Earth station terminal, the frequency of the satellite transmitter also varies with respect to the receiver on the Earth station terminal. If the frequency transmitted by the satellite is f_T , then the received frequency f_R is given by equation 3.18.

$$\left(\frac{f_R - f_T}{f_T} \right) = \left(\frac{\Delta f}{f_T} \right) = \left(\frac{v_T}{v_p} \right) \quad (3.18)$$

Where,

v_T is the component of the satellite transmitter velocity vector directed towards the Earth station receiver

v_p is the phase velocity of light in free space (3×10^8 m/s)

Variation in the Orbital Distance

Effect of Orbital Distance on TDMA Communication

1. **Changing Distance Between Satellite and Earth:**
 - As satellites move in their orbits, the distance between the satellite and the Earth station changes.
 - This change is called range variation.
 -
2. **Impact on Communication Timing:**
 - When a satellite uses TDMA (Time Division Multiple Access), timing is very important.

- TDMA works by assigning time slots to different users to send and receive data.
- If the timing is wrong, users may receive the wrong data or miss it completely.
-
- 3. **Need for Careful Timing:**
 - Because the distance keeps changing, the signal takes more or less time to travel.
 - So, the TDMA frame timing must be adjusted carefully to make sure data arrives at the right time.
 -
- 4. **Which Satellites Are Affected Most:**
 - Low Earth Orbit (LEO) and Medium Earth Orbit (MEO) satellites move quickly and have more variation in distance.

Geostationary satellites stay in one place relative to Earth, so their distance remains mostly constant, and timing is easier to manage.

Solar Eclipse

Satellite Power During Eclipses

1. **Eclipses and Solar Radiation:**
 - Sometimes satellites pass behind a celestial body (like Earth or the Moon).
 - During this time, they don't receive sunlight (solar radiation).
2. **Use of Batteries:**
 - When sunlight is blocked, satellites switch to onboard batteries to keep working.
 - These batteries provide power until sunlight returns.
3. **Battery Preparation by Ground Stations:**
 - Ground control stations prepare the batteries before an eclipse to ensure good performance.
 - This preparation is called battery conditioning.
4. **Battery Conditioning Process:**
 - The batteries are discharged close to their maximum depth (but not fully drained).
 - Then, they are fully recharged just before the eclipse starts.
5. **Thermal Stresses During Eclipse:**
 - As the satellite enters or exits the shadow, there are sudden temperature changes.
 - This creates thermal stress on the satellite's body and components.
6. **Satellite Design for Eclipse Conditions:**
 - Satellites are specially designed to handle:

- Lack of solar power during eclipse.
- Thermal stress caused by rapid heating and cooling.

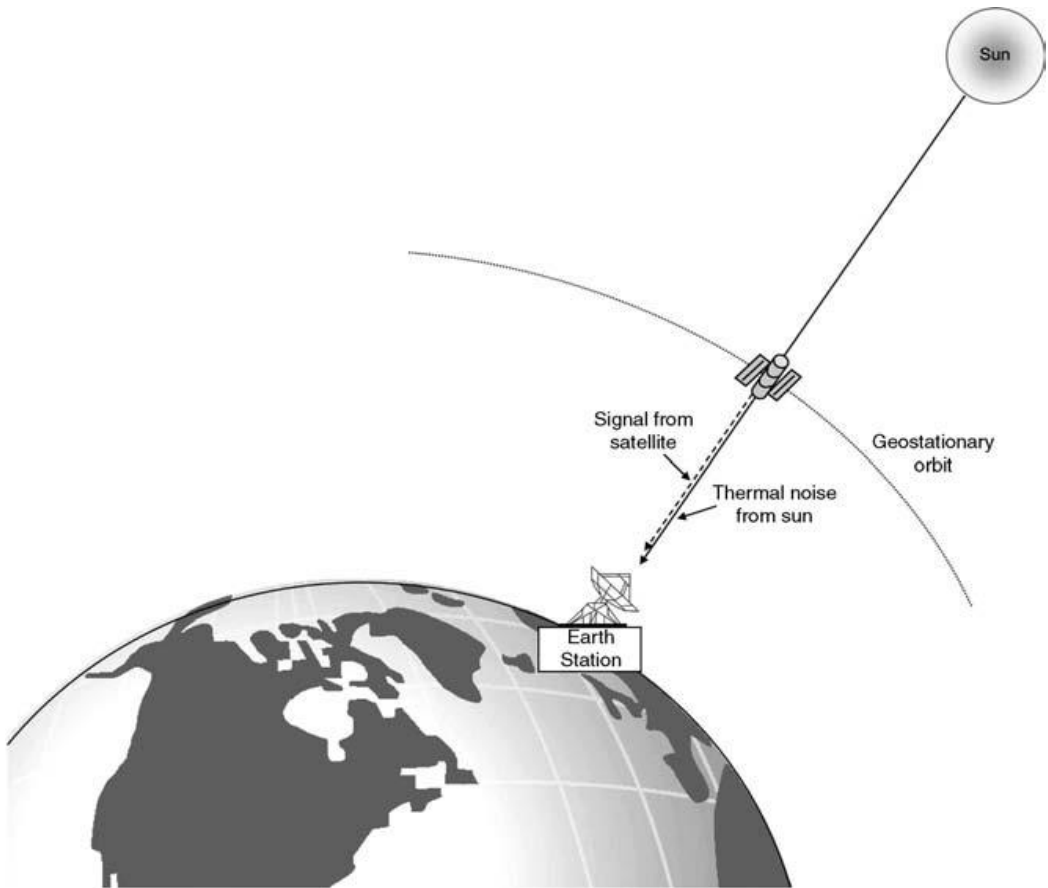
Sun Transit Outrage

Sometimes, the satellite passes directly between the Sun and the Earth. During this time, the Earth station receives signals from both the satellite and the Sun.

Since the Sun also gives off strong microwave radiation, it can interfere with the satellite signal.

If this interference is too strong, it may cause a short loss of signal (called a sun outage).

To avoid problems, satellite traffic may be temporarily moved to another satellite during this period.



Eclipses

An eclipse for a satellite happens when sunlight is blocked from reaching its solar panels because another celestial body (usually the Earth) comes in the way.

The most common cause of this is when the satellite moves into the Earth's shadow, blocking sunlight. This is often referred to as a solar eclipse for the satellite

There are two parts to Earth's shadow:

- The umbra: the dark center part where the satellite gets no sunlight at all — this is a total eclipse.
- The penumbra: the lighter outer part where the satellite gets some sunlight, but not much.

These eclipses happen because the Earth's equator is tilted (about 23.5°) compared to the path it takes around the Sun (called the ecliptic plane). Because of this tilt, the satellite passes through Earth's shadow twice a year:

- Around March (spring equinox)
- Around September (autumn equinox)

During each of these times, the satellite sees an eclipse for about 42 nights. So, that's 84 eclipse nights every year.

During these eclipses, the satellite has to use its batteries since its solar panels can't get sunlight.

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The effect is the worst during the equinoxes and lasts for about 72 minutes. Figure 3.35 further illustrates the phenomenon.

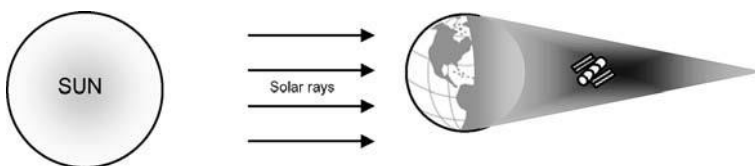


Figure 3.33 Solar eclipse



Figure 3.34 Umbra and penumbra

Hence, the duration of an eclipse increases from zero to about 72 minutes starting 21 days before the equinox and then decreases from 72 minutes to zero during 21 days following the equinox.

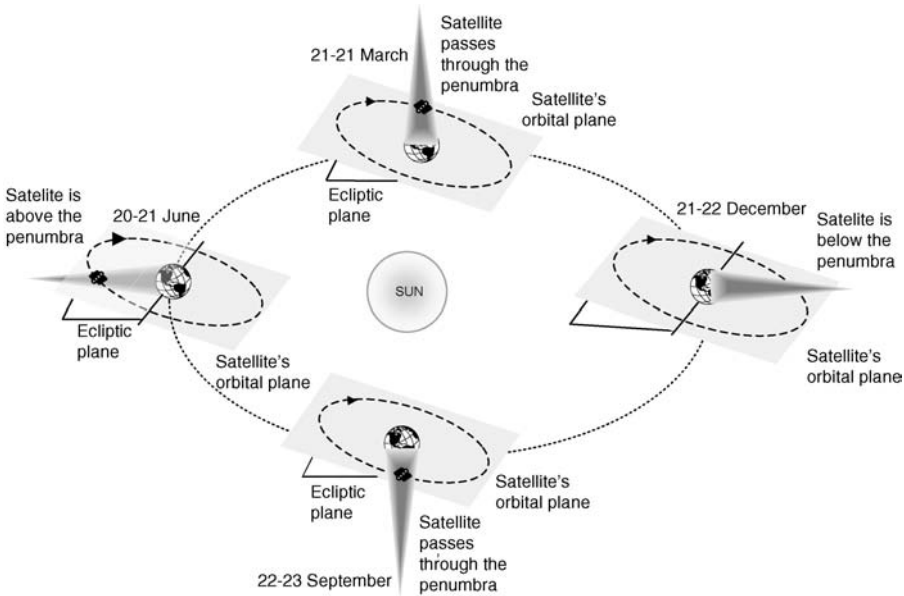


Figure 3.35 Positions of the geostationary satellite during the equinoxes and solstices

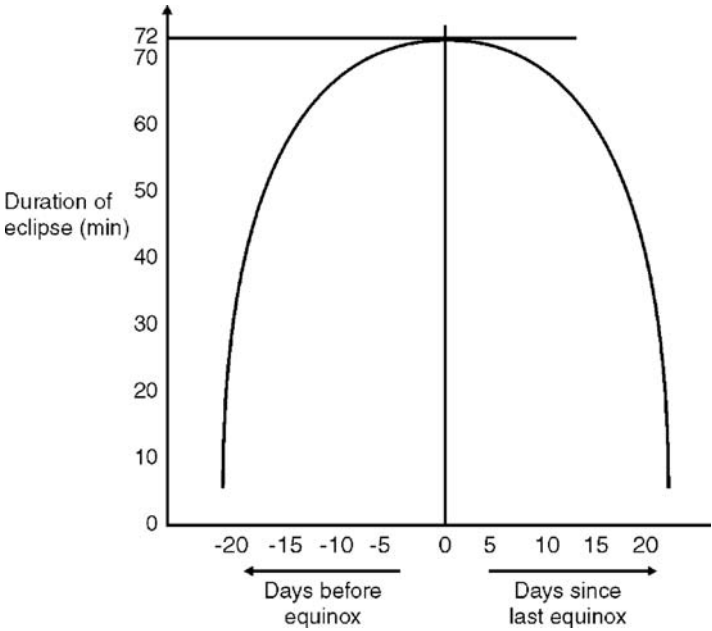


Figure 3.36 Duration of the eclipse before and after the equinox

Another type of eclipse known as the lunar eclipse occurs when the moon's shadow passes across the satellite (Figure 3.37). This is much less common and occurs once in 29 years.

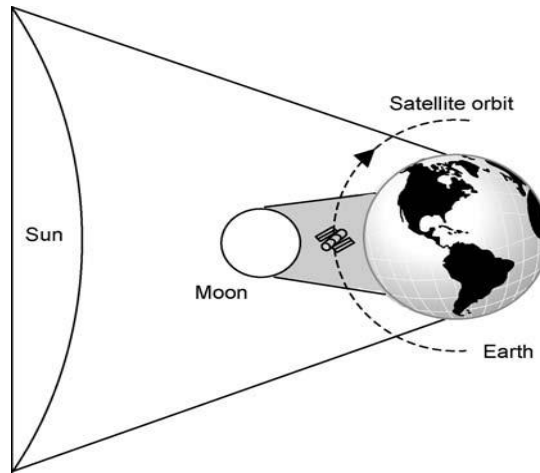


Figure 3.37 Lunar eclipse

While a solar eclipse takes place, the failure of sunlight to reach the satellite interrupts the battery recharging process. The satellite is depleted of its electrical power capacity. It does not significantly affect low power satellites, which can usually continue their operation with back-up power. The high power satellites, however, shut down for all but essential services.

Look Angles of a Satellite

- **Look angles** are the directions in which an **Earth station must point its antenna** to communicate to a satellite.
- These directions are given as two angles:
 - **Azimuth** – The **azimuth** is the **compass direction** from your location to the satellite (or object in the sky).
 - **Elevation** – the angle above the ground (how high to tilt the antenna).
- If the Earth station is inside the **coverage area (footprint)** of a **geostationary satellite**, it can easily connect by **pointing the antenna at the satellite** using these angles.
- To point the antenna correctly, we need to know the azimuth and elevation angles for that location.

- The elevation angle also affects the slant range (the direct distance between the satellite and the Earth station)

i.e. line of sight distance between the Earth station and the satellite.

In order to determine the look angles of a satellite, its precise location should be known. The location of a satellite is very often determined by the position of the sub-satellite point. The sub-satellite point is the location on the surface of the Earth that lies directly between the satellite and the centre of the Earth. To an observer on the sub-satellite point, the satellite will appear to be directly overhead (Figure 3.38).

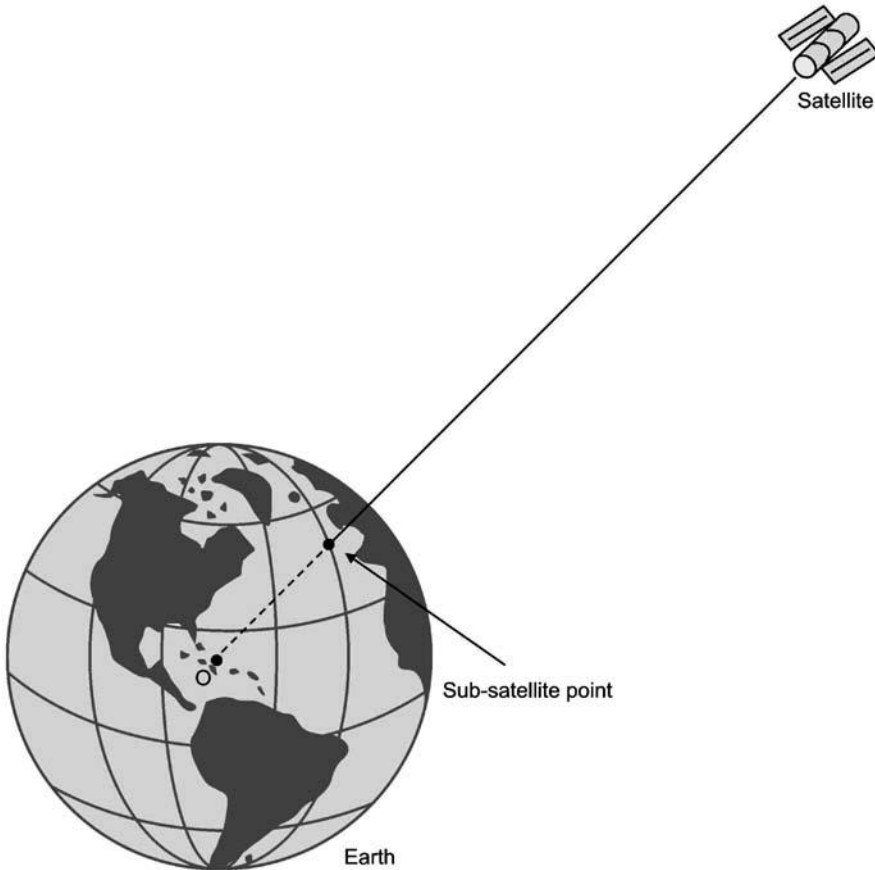


Figure 3.38 Sub-satellite point

Azimuth Angle

The azimuth angle A of an Earth station is defined as the angle produced by the line of intersection of the local horizontal plane and the plane passing through the Earth station, the satellite and the centre of the Earth with the true north (Figure 3.39).

Depending upon the location of the Earth station and the sub-satellite point, the azimuth angle can be computed as follows:

Earth station in the northern hemisphere:

$$A = 180^\circ - A' \quad \text{when the Earth station is to the west of the satellite} \quad (3.19)$$

$$A = 180^\circ + A' \quad \text{when the Earth station is to the east of the satellite} \quad (3.20)$$

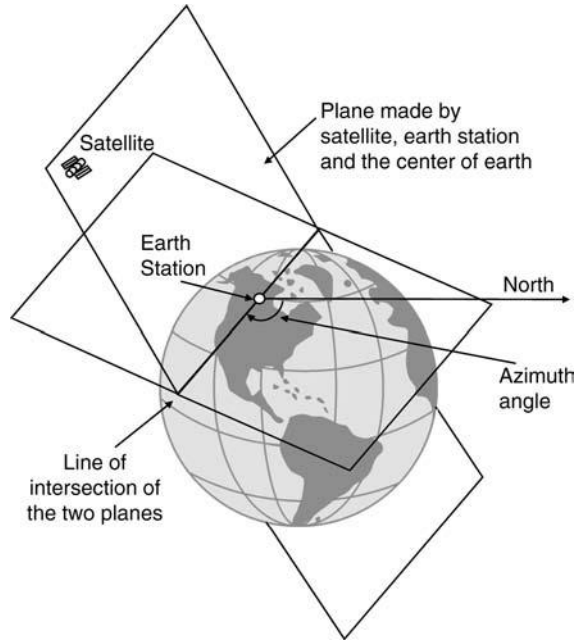


Figure 3.39 Azimuth angle

Earth station in the southern hemisphere:

$$A = A' \dots \quad \text{when the Earth station is to the west of the satellite} \quad (3.21)$$

$$A = 360^\circ - A' \dots \quad \text{when the Earth station is to the east of the satellite} \quad (3.22)$$

where A' can be computed from

$$A' = \tan^{-1} \left(\frac{\tan |\theta_s - \theta_L|}{\sin \theta_l} \right) \quad (3.23)$$

where

θ_s = satellite longitude

θ_L = Earth station longitude

θ_l = Earth station latitude

Elevation Angle

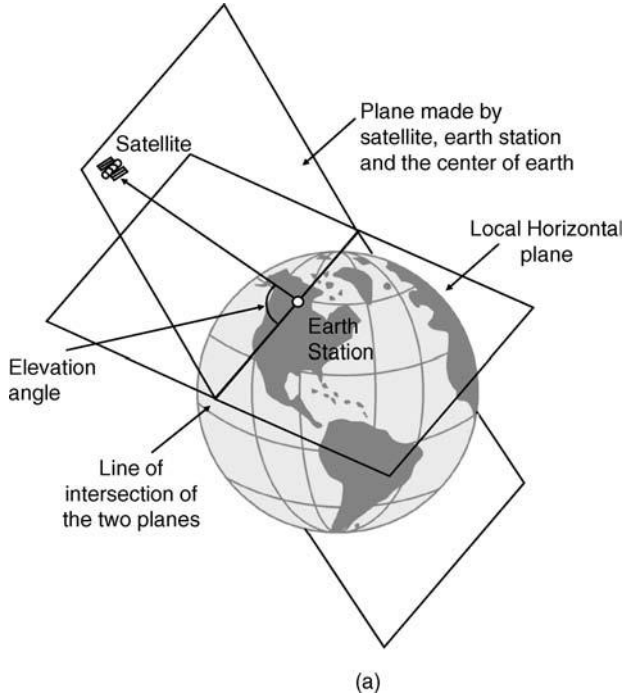
The Earth station elevation angle E is the angle between the line of intersection of the local horizontal plane and the plane passing through the Earth station, the satellite and the centre of the Earth with the line joining the Earth station and the satellite. Figures 3.40 (a) and (b) show the elevation angles for two different satellite and Earth station positions. It can be computed from

$$E = \tan^{-1} \left[\frac{r - R \cos \theta_l \cos |\theta_s - \theta_L|}{R \sin \{\cos^{-1}(\cos \theta_l \cos |\theta_s - \theta_L|)\}} \right] - \cos^{-1}(\cos \theta_l \cos |\theta_s - \theta_L|) \quad (3.24)$$

where

r = orbital radius, R = Earth's radius

θ_s = Satellite longitude, θ_L = Earth station longitude, θ_l = Earth station latitude



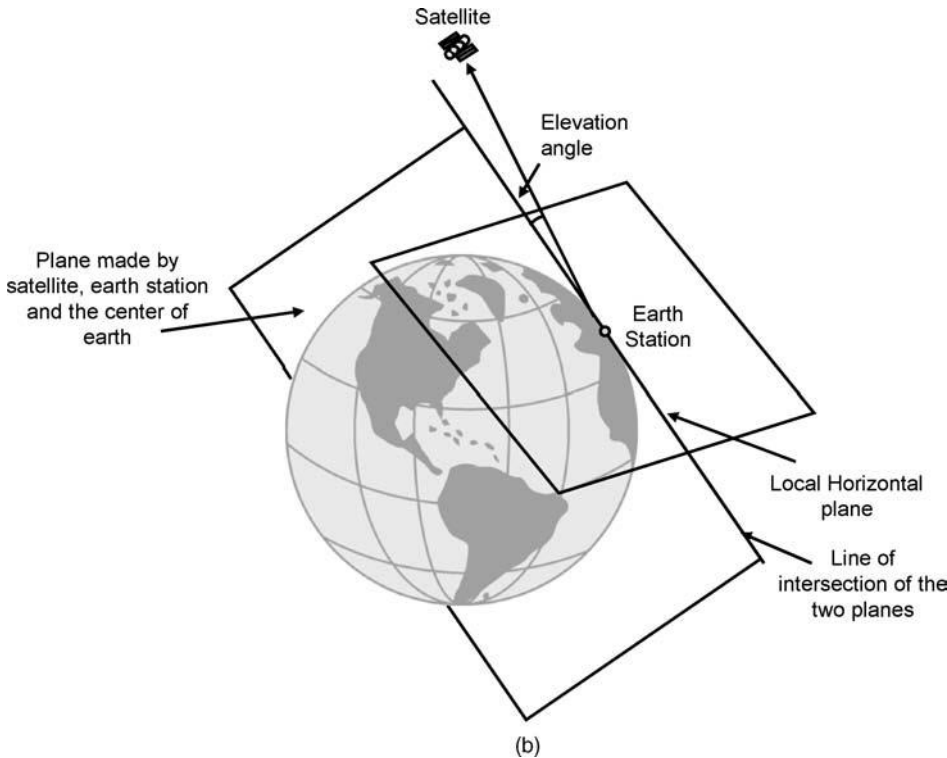


figure 3.40 Earth station elevation angle

Computing the Slant Range

Slant range of a satellite is defined as the range or the distance of the satellite from the Earth station. The elevation angle E , as mentioned earlier, has a direct bearing on the slant range. The smaller the elevation angle of the Earth station, the larger is the slant range and the coverage angle. Refer to Figure 3.41.

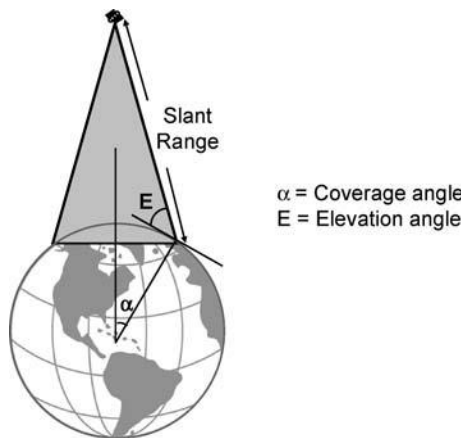


Figure 3.41 Elevation angle, slant range and coverage angle

The slant range can be computed from

$$\text{Slant range } D = \sqrt{R^2 + (R + H)^2 - 2R(R + H) \sin \left[E + \sin^{-1} \left\{ \left(\frac{R}{R + H} \right) \cos E \right\} \right]} \quad (3.25)$$

$$\text{Coverage angle } \alpha = \sin^{-1} \left\{ \left(\frac{R}{R + H} \right) \cos E \right\} \quad (3.26)$$

Where

R = radius of the Earth

E = angle of elevation

H = height of the satellite above the surface of the Earth

It is evident from the above expression that a zero angle of elevation leads to the maximum coverage angle. A larger slant range means a longer propagation delay time and a greater impairment of signal quality, as the signal has to travel a greater distance through the Earth's atmosphere.

Computing the Line-of-Sight Distance between Two Satellites

Refer to Figure 3.42. The line-of-sight distance between two satellites placed in the same circular orbit can be computed from triangle ABC formed by the points of location of two satellites and the centre of the Earth. The line-of-sight distance AB in this case is given by

$$AB = \sqrt{AC^2 + BC^2 - 2 AC BC \cos \theta}$$

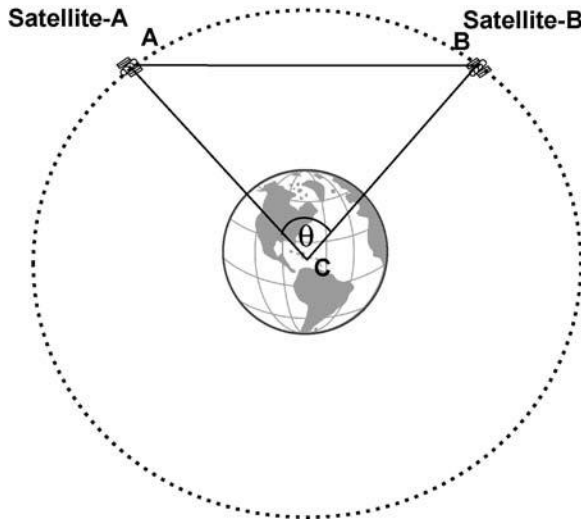


Figure 3.42 Line-of-sight distance between two satellites

The angle ϑ will be the angular separation of the longitudes of the two satellites.

For example, if the two satellites are located at 30°E and 60°E , ϑ would be equal to 30° . If the two locations are 30°W and 60°E , then in that case ϑ would be 90° . The maximum line-of-sight distance between these two satellites occurs when the satellites are placed so that the line

joining the two becomes tangent to the Earth's surface, as shown in Figure 3.43.

In this the case, the maximum line-of-sight distance (AB) equals $OA + OB$, which further equals $2OA$ or $2OB$ as $OA = OB$. If R is the radius of the Earth and H is the height of satellites above the surface of the Earth, then

$$OA = AC \sin \beta = (R + H) \sin \beta \quad (3.28)$$

Now

$$\beta = \cos^{-1} \left(\frac{R}{R + H} \right) \quad (3.29)$$

Therefore

$$OA = (R + H) \sin \left[\cos^{-1} \left(\frac{R}{R + H} \right) \right] \quad (3.30)$$

and

$$\text{Maximum line-of-sight distance} = 2(R + H) \sin \left[\cos^{-1} \left(\frac{R}{R + H} \right) \right] \quad (3.31)$$

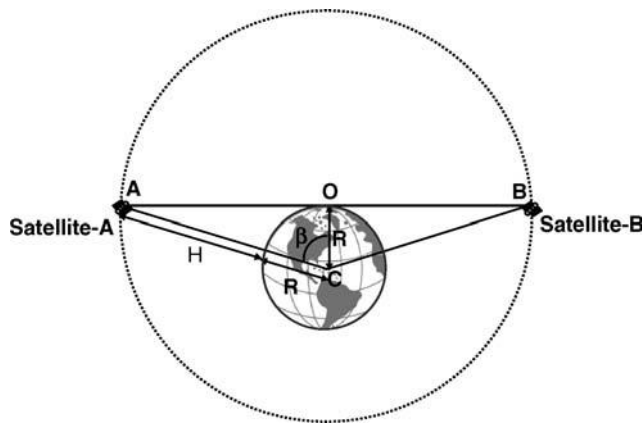


Figure 3.43 Maximum line-of-sight distance between two satellites